21KH0017

REPUBLIC OF THE PHILIPPINES
DEPARTMENT OF PUBLIC WORKS AND HIGHWAYS
REGION X
CAGAYAN DE ORO CITY 1ST DISTRICT ENGINEERING OFFICE
10TH REGIONAL EQUIPMENT SERVICES COMPOUND
BULUA, CAGAYAN DE ORO CITY

C.Y. 2021 PROJECT
DETAILED ENGINEERING DESIGN PLAN FOR
CONCRETING OF ROAD WITH DRAINAGE MANTICAO DRIVE PHASE II,
STA. 0+000.000 - STA. 0+263.875
BRGY. CANITOAN, CAGAYAN DE ORO CITY, MISAMIS ORIENTAL
NET LENGTH = 263.875 m

SUBMITTED:
JOSE MARCOS S. DACANAY
CHIEF, PLANNING AND DESIGN SECTION
DATE:

RECOMMENDED:
EVELYN A. TIRADO
ASSISTANT DISTRICT ENGINEER
DATE:

APPROVED:
VIRGINIA G. BATOCAIL
DISTRICT ENGINEER
DATE:

TABLE OF CONTENTS

DRAWING TITLE		SET NO.	SHEET NO.
TITLE SHEET			
TABLE OF CONTENTS			1
G. GENERAL DRAWING			
KEY MAP, MINDANAO MAP, CAGAYAN DE ORO MAP	(1/15)	G-1	2
LOCATION AND DISPOSAL MAP	(2/15)	G-2	3
MATERIALS SOURCE MAP	(3/15)	G-3	4
PORTLAND CEMENT CONCRETE PAVEMENT DETAILS	(4/15)	G-4	5
PAVEMENT DESIGN ANALYSIS	(5/15)	G-5	6
GENERAL NOTES	(6/15)	G-6	7
GEOMETRIC DESIGN STANDARD DRAWINGS	(7/15)	G-7	8
GEOMETRIC DESIGN STANDARD DRAWINGS	(8/15)	G-8	9
CONSTRUCTION REQUIREMENTS	(9/15)	G-9	10
CONSTRUCTION REQUIREMENTS	(10/15)	G-10	11
CONSTRUCTION REQUIREMENTS	(11/15)	G-11	12
ROADWAY STREET LIGHTING	(12/15)	G-12	13
TYPICAL TRAFFIC SIGNAGE LAYOUT	(13/15)	G-13	14
TRAFFIC SIGNS AND HAZARD MARKERS	(14/15)	G-14	15
PROJECT BILLBOARD	(15/15)	G-15	16
P. PLAN AND PROFILE			
STA. 0 + 000.000 - STA. 0 + 263.875	(1/2)	P-1	17
JUNCTION DETAILS	(2/2)	P-2	18
S. TYPICAL SECTIONS, DETAILS AND QUANTITIES			
TYPICAL ROADWAY SECTIONS & RCPC DETAILS	(1/3)	S-1	19
TYPICAL SECTIONS OF REINFORCED CONCRETE CANALS, STEEL HANDLE DETAIL	(2/3)	S-2	20
SUMMARY OF QUANTITIES, GRADING QUANTITIES, PAVING QUANTITIES, SCHEDULE OF DRAINAGE	(3/3)	S-3	21

DRAWING TITLE		SET NO.	SHEET NO.
C. CROSS SECTIONS			
STA. 0+000.000 - STA. 0 +060.000	(1/4)	C-1	22
STA. 0+080.000 - STA. 0 +140.000	(2/4)	C-2	23
STA. 0+160.000 - STA. 0 +200.000	(3/4)	C-3	24
STA. 0+220.000 - STA. 0 +263.875	(4/4)	C-4	25

REPUBLIC OF THE PHILIPPINES
DEPARTMENT OF PUBLIC WORKS AND HIGHWAYS
REGION X
CAGAYAN DE ORO CITY 1ST DISTRICT
ENGINEERING OFFICE
10TH REGIONAL EQUIPMENT SERVICES COMPOUND
BULUA, CAGAYAN DE ORO CITY

PROJECT NAME AND LOCATION:
CONCRETING OF ROAD WITH DRAINAGE
MANTICAO DRIVE PHASE II, BRGY.
CANITOAN, CAGAYAN DE ORO CITY,
MISAMIS ORIENTAL

SHEET CONTENTS:
TABLE OF CONTENTS

SURVEYED:
[illegible] M. ABENIR
ENGINEER II

DRAFTED / PREPARED:
[illegible]

REVIEWED:
[illegible] B. DAGALA
ENGINEER II
DATE

SUBMITTED:
JO[illegible] S. DACA[illegible]
[illegible] AND DESIGN SECTION
DATE

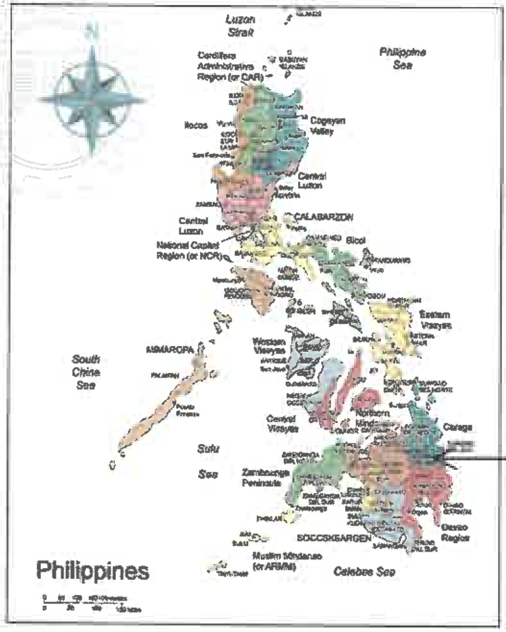
RECOMMENDED:
E[illegible] TIRADO
ASSISTANT DISTRICT ENGINEER
DATE

APPROVED:
VI[illegible]
DISTRICT ENGINEER
DATE

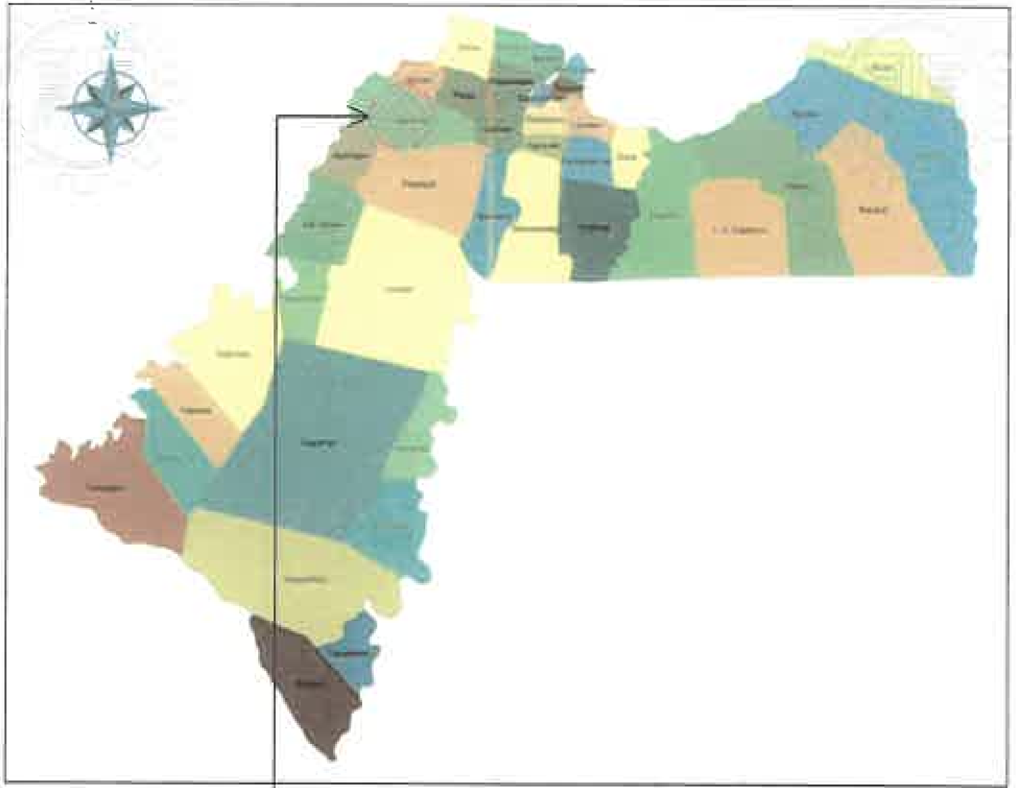
SET NO.: 1 / 1

SHEET NO.: 1 / 25

G. GENERAL DRAWINGS


Luzon Strait
Cordillera Administrative Region (or CAR)
Philippine Sea
Ilocos
Cagayan Valley
Central Luzon
Central Luzon
CALABARZON
National Capital Region (or NCR)
Bicol
MIMAROPA
Eastern Visayas
Western Visayas
South China Sea
Central Visayas
Northern Mindanao
Caraga
Sulu Sea
Zamboanga Peninsula
Davao Region
SOCCSKSARGEN
Muslim Mindanao (or ARMM)
Celebes Sea
Philippines

① G1 KEY MAP
NOT TO SCALE

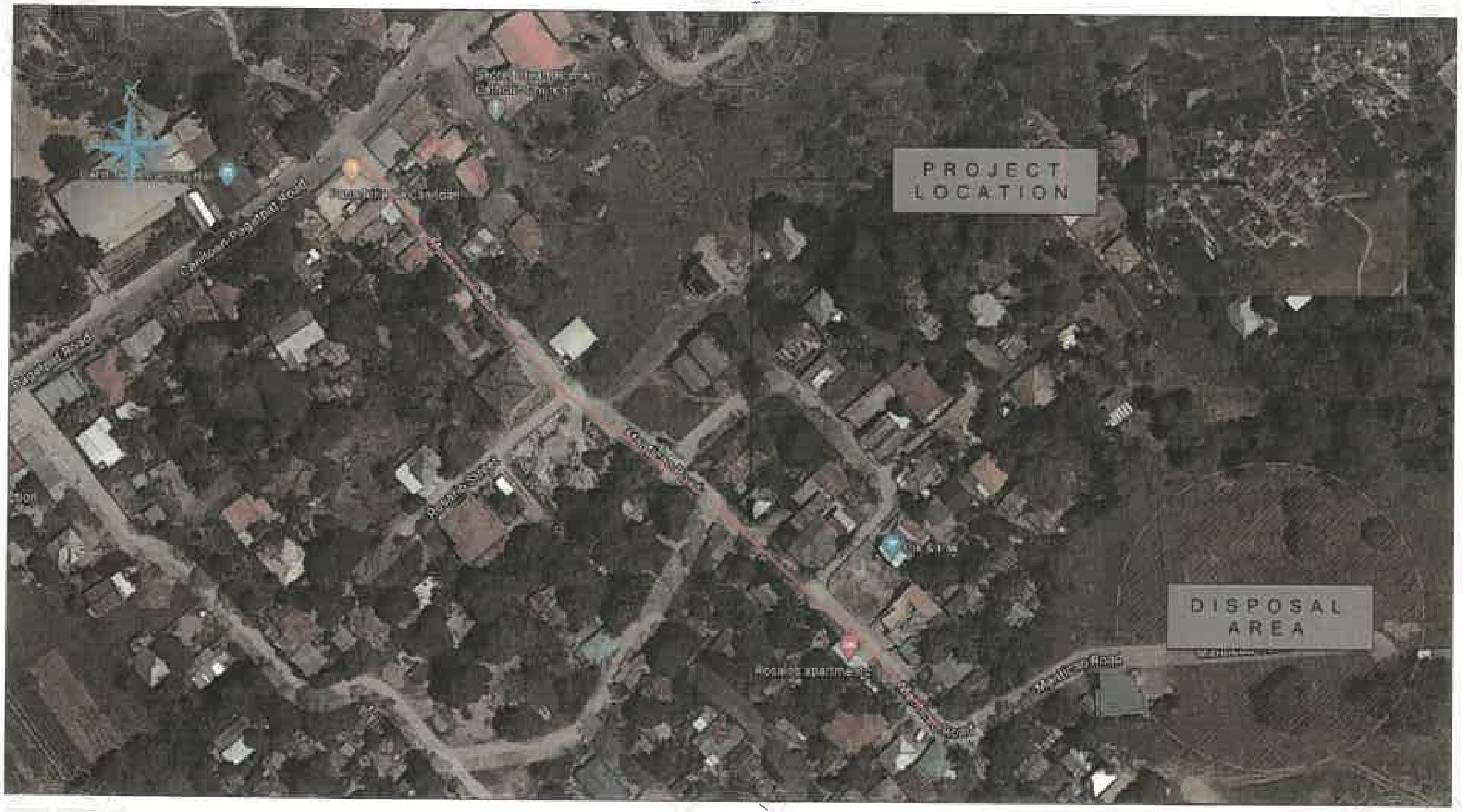


③ G1 CAGAYAN DE ORO MAP
NOT TO SCALE

THIS SITE

② G1 MINDANAO MAP

REPUBLIC OF THE PHILIPPINES DEPARTMENT OF PUBLIC WORKS AND HIGHWAYS REGION X CAGAYAN DE ORO CITY 1ST DISTRICT ENGINEERING OFFICE 10TH REGIONAL EQUIPMENT SERVICES COMPOUND BULUA, CAGAYAN DE ORO CITY	PROJECT NAME AND LOCATION: CONCRETING OF ROAD WITH DRAINAGE MANTICAO DRIVE PHASE II, BRGY. CANITOAN, CAGAYAN DE ORO CITY, MISAMIS ORIENTAL	SHEET CONTENTS: KEY MAP, MINDANAO MAP, CAGAYAN DE ORO MAP	SURVEYED: ENGINEER II DRAFTED:	B. DAGALA ENGINEER II DATE	S. DACANAY AND DESIGN SECTION DATE	RECOMMENDED: DISTRICT ENGINEER DATE	APPROVED: DATE	SET NO.: G 1 15 SHEET NO.: 2 / 25



LOCATION AND DISPOSAL MAP

1 / G 2

NOT TO SCALE

REPUBLIC OF THE PHILIPPINES
DEPARTMENT OF PUBLIC WORKS AND HIGHWAYS
REGION X
CAGAYAN DE ORO CITY 1ST DISTRICT
ENGINEERING OFFICE
10TH REGIONAL EQUIPMENT SERVICES COMPOUND
BULUA, CAGAYAN DE ORO CITY

PROJECT NAME AND LOCATION:
CONCRETING OF ROAD WITH DRAINAGE MANTICAO DRIVE PHASE II, BRGY. CANITOAN, CAGAYAN DE ORO CITY, MISAMIS ORIENTAL

SHEET CONTENTS:
LOCATION MAP AND DISPOSAL MAP

SURVEYED:
[illegible]
ENGINEER II

DRAFTED:
[illegible]

REVIEWED:
[illegible] B. DATGALA
ENGINEER II
DATE

SUBMITTED:
JOSE MARCOS E. DACANAY
CHIEF, PLANNING AND DESIGN SECTION
DATE

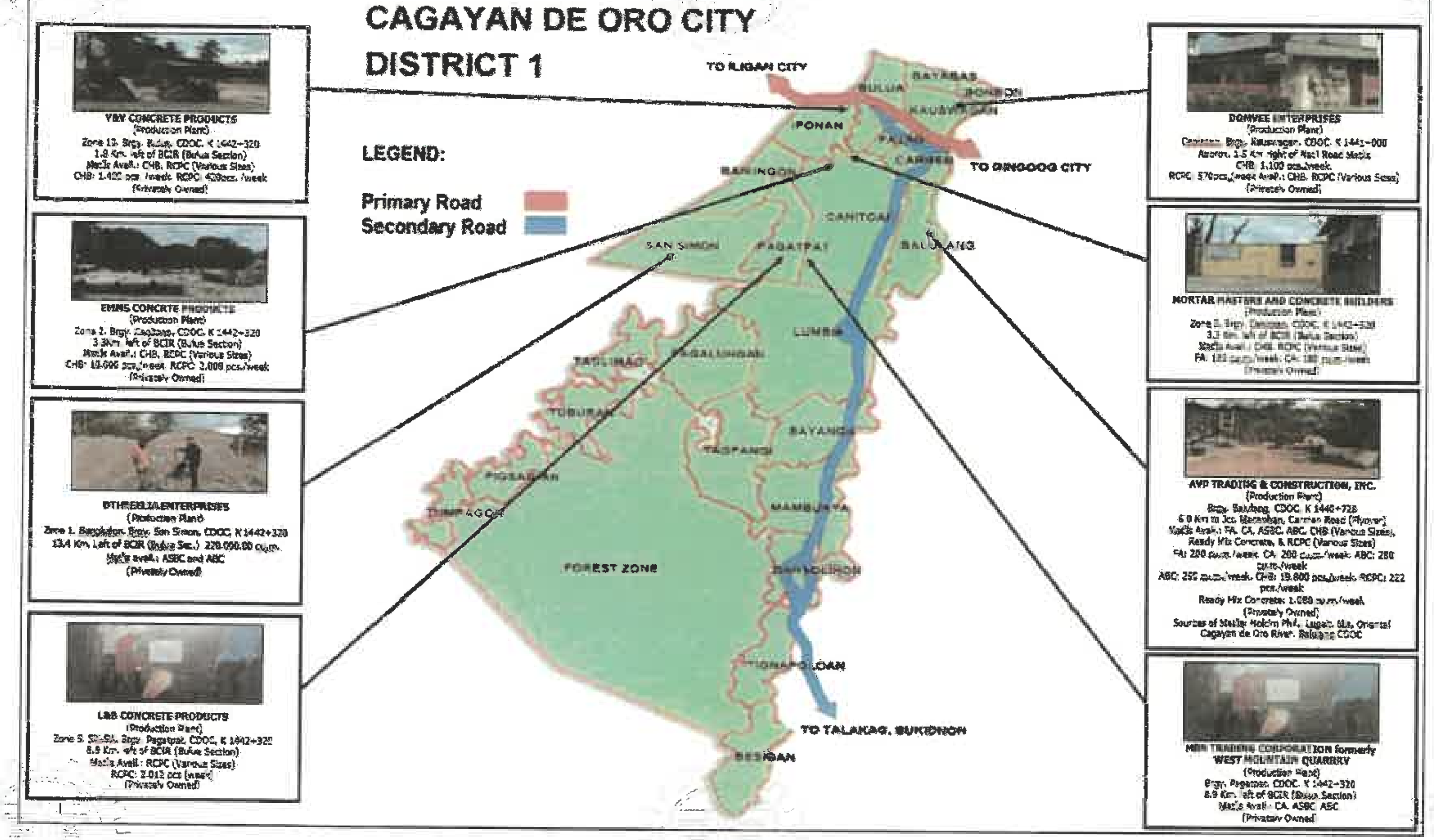
RECOMMENDED:
EVELYN A. TIRADO
ASSISTANT DISTRICT ENGINEER
DATE

APPROVED:
[illegible] SATUCAIL
DISTRICT ENGINEER
DATE

SET NO.: G 2 15

SHEET NO.: 3 25

MATERIALS SOURCE MAP


CAGAYAN DE ORO CITY
DISTRICT 1
LEGEND:
Primary Road 🟥
Secondary Road 🟦
TO ILIGAN CITY
BAYABAS
BONBON
BULUA
KAUSWAGAN
PONAN
PATAG
CARMEN
TO GINGOOG CITY
BANINGON
CANITOAN
BALULANG
SAN SIMON
PAGATPAT
LUMBIA
TAGLIMAO
PAGALUNGAN
TUBURAN
BAYANGA
TAGPANGI
PIGSAG-AN
MAMBUAYA
DANSOLIHON
TUMPAGON
FOREST ZONE
TIGNAPOLOAN
TO TALAKAG, BUKIDNON
BESIGAN

VBY CONCRETE PRODUCTS
(Production Plant)
Zone 12, Brgy. Bulua, CDOC, K 1442+320
1.8 Km. left of BCIR (Bulua Section)
Mat'ls Avail.: CHB, RCPC (Various Sizes)
CHB: 1,422 pcs. /week, RCPC: 420pcs. /week
(Privately Owned)

EMMS CONCRETE PRODUCTS
(Production Plant)
Zone 2, Brgy. Canitoan, CDOC, K 1442+320
3.3Km. left of BCIR (Bulua Section)
Mat'ls Avail.: CHB, RCPC (Various Sizes)
CHB: 10,000 pcs./week, RCPC 2,000 pcs./week
(Privately Owned)

DTHREELIA ENTERPRISES
(Production Plant)
Zone 1, Bangkalan, Brgy. San Simon, CDOC, K 1442+320
13.4 Km. Left of BCIR (Bulua Sec.) 220,000.00 cu.m.
Mat'ls avail.: ASBC and ABC
(Privately Owned)

L&B CONCRETE PRODUCTS
(Production Plant)
Zone 5, Sili-Sili, Brgy. Pagatpat, CDOC, K 1442+320
8.9 Km. left of BCIR (Bulua Section)
Mat'ls Avail.: RCPC (Various Sizes)
RCPC: 2,012 pcs./week
(Privately Owned)

DOMVEE ENTERPRISES
(Production Plant)
Cagahan, Brgy. Kauswagan, CDOC, K 1441-000
Aurora, 1.5 Km right of Nat'l Road Mat'ls
CHB: 1,100 pcs./week.
RCPC: 570pcs./week Avail.: CHB, RCPC (Various Sizes)
(Privately Owned)

MORTAR MASTERS AND CONCRETE BUILDERS
(Production Plant)
Zone 2, Brgy. Canitoan, CDOC, K 1442+320
3.3 Km. left of BCIR (Bulua Section)
Mat'ls Avail.: CHB, RCPC (Various Sizes)
FA: 180 cu.m./week, CA: 180 cu.m./week
(Privately Owned)

AVP TRADING & CONSTRUCTION, INC.
(Production Plant)
Brgy. Balulang, CDOC, K 1440+728
6.0 Km to Jct. Macanhan, Carmen Road (Flyover)
Mat'ls Avail.: FA, CA, ASBC, ABC, CHB (Various Sizes), Ready Mix Concrete, & RCPC (Various Sizes)
FA: 200 cu.m./week, CA: 200 cu.m./week, ABC: 280 cu.m./week
ABC: 250 cu.m./week, CHB: 19,800 pcs./week, RCPC: 222 pcs./week
Ready Mix Concrete: 1,080 cu.m./week
(Privately Owned)
Sources of Mat'ls: Holcim Phil., Lugait, Mis. Oriental
Cagayan de Oro River, Balulang CDOC

MBN TRADING CORPORATION formerly WEST MOUNTAIN QUARRY
(Production Plant)
Brgy. Pagatpat, CDOC, K 1442+320
8.9 Km. left of BCIR (Bulua Section)
Mat'ls Avail.: CA, ASBC, ABC
(Privately Owned)

MATERIAL SOURCE MAP

1 / G 3

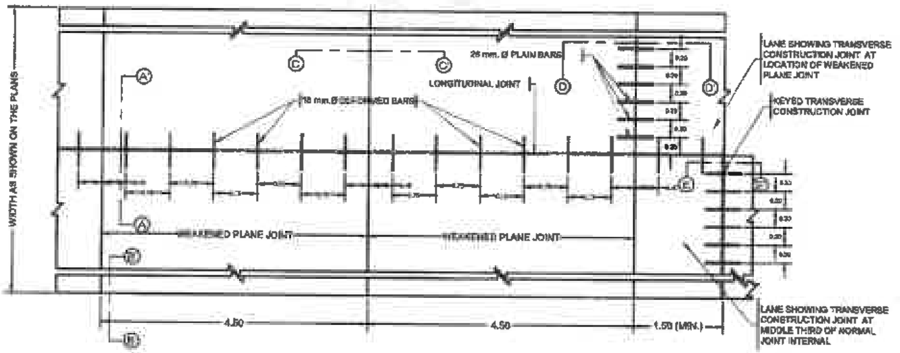
NOT TO SCALE

REPUBLIC OF THE PHILIPPINES DEPARTMENT OF PUBLIC WORKS AND HIGHWAYS REGION X CAGAYAN DE ORO CITY 1ST DISTRICT ENGINEERING OFFICE 10TH REGIONAL EQUIPMENT SERVICES COMPOUND BULUA, CAGAYAN DE ORO CITY	PROJECT NAME AND LOCATION: CONCRETING OF ROAD WITH DRAINAGE MANTICAO DRIVE PHASE II, BRGY. CANITOAN, CAGAYAN DE ORO CITY MISAMIS ORIENTAL	SHEET CONTENTS: MATERIAL SOURCE MAP	SURVEYED: [illegible] ENGINEER II DRAFTED BY: JOHNE GRACE [illegible] ENGINEER II	REVIEWED: [illegible] B. DAGALA ENGINEER V DATE	SUBMITTED: [illegible] S. DACAMAY AND DESIGN SECTION DATE	RECOMMENDED: [illegible] N. TIRADO ASSISTANT DISTRICT ENGINEER DATE	APPROVED: VIRGINIO [illegible] DISTRICT ENGINEER DATE	SET NO.: 6 3 | 16
								SHEET NO.: 4 25

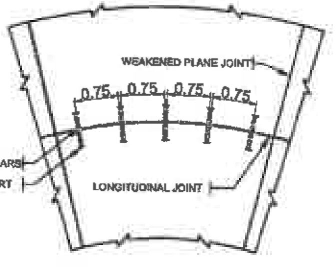
PORTLAND CEMENT CONCRETE PAVEMENT DETAILS

NOTE:

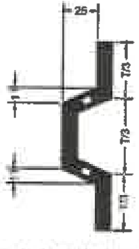
1. MATERIALS AND WORKMANSHIP SHALL CONFORM WITH THE "DPWH STANDARD SPECIFICATIONS FOR HIGHWAYS, BRIDGES AND AIRPORTS, 2013".
2. CONSTRUCTION (CONTACT) JOINTS ARE FORMED WHEN CONCRETE ON ONE SIDE OF THE JOINT IS POURED AHEAD AND ALLOWED TO SET BEFORE POURING ON THE OTHER SIDE. NO CONSTRUCTION JOINT SHALL BE PLACED WITHIN 1.50M. FROM THE WEAKENED PLANE JOINT.
3. TRANSVERSE CONSTRUCTION (CONTACT JOINT) SHALL BE PROVIDED AT THE END OF ANY RUN WHERE LAYING OF CONCRETE HAS BEEN STOPPED FOR THIRTY (30) MINUTES OR LONGER.
4. TRANSVERSE CONSTRUCTION JOINTS WHICH OCCUR AT LOCATION OF WEAKENED PLANE JOINTS SHOULD BE BUTT JOINTS WITH DOWELS IF THE JOINT OCCURS IN THE MIDDLE THIRD OF THE WEAKENED JOINT INTERVALS (1.50 TO 3.00 m.) IT SHOULD BEKEY - JOINT WITH TIE BARS.
5. AT CONSTRUCTION JOINTS (LONGITUDINAL OR TRANSVERSE), CARE SHOULD BE TAKEN SO THAT NO CONCRETE FROM THE LAST SLAB PLACED OVERHANGS ANY PORTION OF THE FIRST SLAB.
6. TIE BARS SHOULD BE DEFORMED STEEL BARS. ALL DOWEL BARS SHALL BE SMOOTH ROUND STEEL BAR FREE FROM RUST AND OTHER DEFECTS (REFER TO CLAUSE 3: 1.3.12. THE STANDARD SPECIFICATION FOR ROADS AND BRIDGES 2013.).
7. TYPE OF WEAKENED PLANE JOINT TO BE USED SHALL BE AS SPECIFIED IN THE PLANS AND ONLY ONE TYPE SHALL BE USED FOR THE WHOLE PROJECT.
8. MATERIAL FOR THE METAL SIDE FORM SHALL BE BRAND NEW SHEET METAL GAUGE No.16 OF BLACK IRON FREE FROM RUST AND LINKS.
9. AT LEAST SIX (6) SUCCESSIVE DOWELED BUTT JOINTS AT NORMAL JOINT SPACING SHALL BE PROVIDED BEFORE OR AFTER AN EXPANSION JOINT.
10. THE GROOVE OR CRACK ABOVE JOINTS (LONGITUDINAL OR TRANSVERSE) SHALL BE SEALED WITH 30-60 PENETRATION ASPHALT SEAL OR COLD APPLIED LIQUID RUBBER COMPOUND AFTER THE CONCRETE HARDENS AND BEFORE OPENING THE PAVEMENT TO TRAFFIC. PENETRATION ASPHALT SEAL ON CONCRETE PAVEMENT, JOINTS SHALL BE POURED IN SUCH MANNER THAT SPILLING WILL BE ELIMINATED / PREVENTED. THUS, PROVIDING A SMOOTH LEVELING / RIDING SURFACE.
11. ALL TRANSVERSE JOINTS, EXCEPT CONSTRUCTION JOINT SHALL BE CONTINUOUS FROM EDGE TO EDGE.
12. ALL LONGITUDINAL JOINTS SHALL MEET AT INTERSECTIONS WITH NO GAPS OR OFFSET.
13. AVOID STOPPAGE OF FORMWORKS ALONG CURVES.


WIDTH AS SHOWN ON THE PLANS
A
A'
C
C'
D
D'
E
E'
B
B'
26 mm. Ø PLAIN BARS
LONGITUDINAL JOINT
16 mm. Ø DEFORMED BARS
WEAKENED PLANE JOINT
WEAKENED PLANE JOINT
4.50
4.50
1.50 (MIN.)
LANE SHOWING TRANSVERSE CONSTRUCTION JOINT AT LOCATION OF WEAKENED PLANE JOINT
KEYED TRANSVERSE CONSTRUCTION JOINT
LANE SHOWING TRANSVERSE CONSTRUCTION JOINT AT MIDDLE THIRD OF NORMAL JOINT INTERNAL

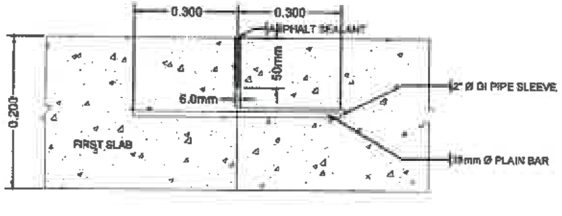
PORTLAND CEMENT CONCRETE PAVEMENT DETAIL
SCALE : NTS


WEAKENED PLANE JOINT
0.75
0.75
0.75
0.75
16mm.Ø DEFORMED BARS
BAR SUPPORT
LONGITUDINAL JOINT

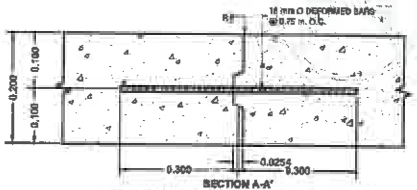
TRANSVERSE CONSTRUCTION JOINT / WEAKENED JOINT
SCALE : NTS



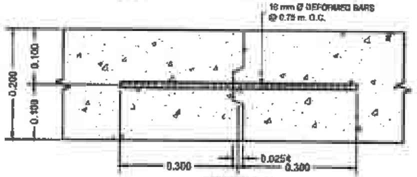
METAL SIDE FORM (KEYED JOINT) DETAIL
SCALE : NTS


0.300
0.300
ASPHALT SEALANT
50mm
6.0mm
0.200
FIRST SLAB
2" Ø GI PIPE SLEEVE
16mm Ø PLAIN BAR

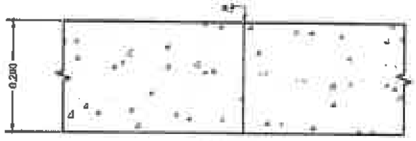
PIPE SLEEVES DETAIL
SCALE : NTS


16 mm Ø DEFORMED BARS @ 0.75 m. O.C.
0.200
0.100
0.100
0.0254
0.300
0.300

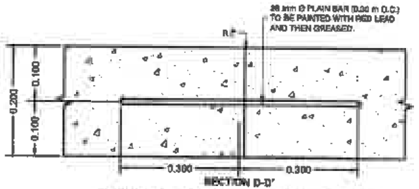
SECTION A-A'
LONGITUDINAL CONSTRUCTION OR CONTACT JOINT
(TO BE USED FOR HALF WIDTH OR LANE AT A TIME CONSTRUCTION)


16 mm Ø DEFORMED BARS @ 0.75 m. O.C.
0.200
0.100
0.100
0.0254
0.300
0.300

SECTION B-B'
LONGITUDINAL CENTER METAL PLATE JOINT
(TO BE USED FOR FULL WIDTH OR SIMULTANEOUS CONSTRUCTION OF TWO OR MORE LANES)



SECTION C-C'
SCALE NTS


26 mm Ø PLAIN BAR (0.30 m O.C.) TO BE PAINTED WITH RED LEAD AND THEN GREASED.
R
0.200
0.100
0.100
0.300
0.300

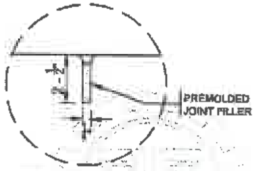
SECTION D-D'
TRANSVERSE CONTRACTION JOINT/ WEAKENED JOINT
(TO BE PLACED ONLY AT LOCATION WEAKENED PLANE)

SPACING OF DOWEL BAR (Diameter 36mm/L=600mm)	
SLAB DEPTH, (T) (mm)	SPACING (mm)
280	300
290	285
300	270
310	255
320	235
330	220
340	205

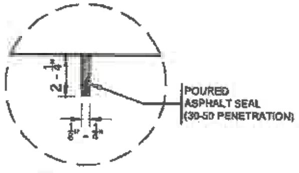
NOTE:
DIAMETER AND SPACING OF PLAIN DOWEL BARS MAY BE MODIFIED AS LONG AS THE EQUIVALENT STEEL AREA IS MAINTAINED/ INCREASED.

TABLE FOR SPACING OF TIE BARS (L=600mm)		
SLAB THICKNESS (mm)	SPACING, S1 (mm)	
	12mm Ø	16mm Ø
230	600	750
240	600	750
250	600	750
260	500	750
270	500	750
280	500	750
290	500	750
300	500	750
310	400	750
320	400	750
330	400	750
340	400	750

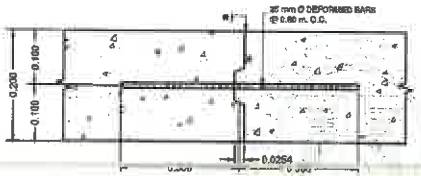
Based on AASHTO Pavement Design 1993


PREMOLDED JOINT FILLER

PREMOLDED STRIP TYPE
SCALE : NTS


POURED ASPHALT SEAL (30-50 PENETRATION)

SAWED GROOVE TYPE
SCALE : NTS


25 mm Ø DEFORMED BARS @ 0.60 m. O.C.
0.200
0.100
0.100
0.0254

SECTION E-E'
KEYED TRANSVERSE CONSTRUCTION OR CONTACT JOINT
(TO BE PLACED ONLY IN MIDDLE THIRD OF NORMAL JOINT INTERVAL)

REPUBLIC OF THE PHILIPPINES
DEPARTMENT OF PUBLIC WORKS AND HIGHWAYS
REGION X
CAGAYAN DE ORO CITY 1ST DISTRICT
ENGINEERING OFFICE
10TH REGIONAL EQUIPMENT SERVICES COMPOUND
BULUA, CAGAYAN DE ORO CITY

PROJECT NAME AND LOCATION:
CONCRETING OF ROAD WITH DRAINAGE
MANTICAO DRIVE PHASE II, BRGY.
CANITOAN, CAGAYAN DE ORO CITY,
MISAMIS ORIENTAL

SHEET CONTENTS:
PORTLAND CEMENT CONCRETE
PAVEMENT DETAILS

SURVEYED:
ENGINEER II

DRAFTED / CHECKED:
JONNIE GRACE P. CAMPOS

REVIEWED:
ANDERT B. DAGALA
ENGINEER II
DATE

SUBMITTED:
JOSE MARCELINO DACANAY
AND DESIGN SECTION
DATE

RECOMMENDED:
EVELYN A. TIRADO
ASSISTANT DISTRICT ENGINEER
DATE

APPROVED:
VIRGINIA C. SAYSON
DISTRICT ENGINEER
DATE

SET NO.: G 4 15

SHEET NO.: 5 25

PAVEMENT DESIGN ANALYSIS

PAVEMENT DESIGN CALCULATION SHEET (TRIAL)

Name of Project :	Location :	Date :
Concreting of Road with Drainage, Manticao Drive Phase II **Brgy. Canitoan, Cagayan de Oro City, Misamis Oriental**	Cagayan de Oro City Length : 0.264 km km 0.264 km	Allocation ('000): PhP 3,000,000.00 PhP PhP 3,000,000.00
Type fo Pavement	Design Life :	Source of Data:
RIGID PAVEMENT	20 Years	

RIGID PAVEMENT DESIGN

DESIGN INPUTS :

Life Design	=	20.00	Years
TGR	=	2.00 %	Assumed Traffic Growth
R	=	85%	
zR	=	- 1.037	1,511,083
so	=	0.35	Overall Standard Deviation
W18	=	1.511	x 106 (18 KIP ESAL)
pt	=	2.50 /3	0.67
DPSI	=	2.00	Present Serviceability Index
S'c	=	761	psi
Cd	=	1.00	
J	=	3.80	
Ec	=	3.605	x 105 psi
k	=	130	psi

Initial D = 8.00000 inches or 20.32 cm
D+1 = 9.000 inches

DESIGN CONDITIONS :

Design Period =	20	Years
Actual Loading =	100 %	(including overloaded trucks)
Design ESAL =	1,511,083	One direction
Conc. Strength =	650	psi @ 28 Days

Design CBR: = Mean CBR - ¾S

where: $S = \sqrt{((\ -\ )^2/(n-1))}$

Mean CBR = 12.9000

CBR = 12.900

16,200.00

Resilient Modulus of Subgrade:

MR = CBR X 1500

MR = 19,350

Composite Modulus of Subgrade:

k∞ = 530

Corrected k, (from the graphs):

kcor. = 130

DESIGN EQUATION :

$$\log_{10}W_{18} = Z_R \times S_0 + 7.35 \times \log_{10}(D+1) - 0.06 + \frac{\log_{10}[DPSI/(4.5-1.5)]}{1.00+[(1.624\times10^{7})/(D+1)^{8.46}]} + (4.22-0.32) \times \log_{10}\frac{[S'_c \times C_d (D^{0.75} - 1.132)]}{[215.63 \times J (D^{0.75}-(18.42/(E_c/k)^{0.25}))]}$$

Trial & Error Method using the Design Equation :

6.1793 = 6.1793

RESULT :

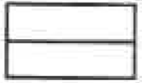
PCCP Thickness Determination :

D1 = 8.000 inches OR

D1 = 203.200 mm

∴ Use D1 = 200 mm

PAVEMENT STRUCTURE



PCC Thickness, TPCC = 200 mm

Subbase, Item 200 = 100 mm

Subgrade CBR = 12.900

MR, resilient modulus = 19,350

TRAFFIC AND CBR DATA

TRAFFIC ANALYSIS :

Assumed Traffic Growth Rate = 2%

		Initial AADT, 2010		Opening Year AADT, 2011
Large Bus	=	0		0
Rigid Truck, 2 Axles	=	50	20	52
Rigid Truck, 3 Axles	=	25	3	26
Truck, Semi-Trailer, 3 - 4 Axles	=	0		0
Truck, Semi-Trailer, 5 Axles	=	0		0
Truck,Trailer, > 4 Axles	=	5		6
Truck,Trailer, >5 Axles	=	0		0
		80 (both direction)		84 (both direction)

Basic Formula :

Design Traffic = AADT $\times \left(\frac{(1+0.04)^{20}-1}{0.04}\right) \times 365$

10,858.9957

Vehicle Class	Design Traffic	Equivalence Factors	CESAL
Large Bus	0	0.56	0
Rigid Truck, 2 Axles	543,450	1.38	749,961
Rigid Truck, 3 Axles	271,725	7.01	1,904,792
Truck, Semi-Trailer, 3 - 4 Axles	0	3.27	0
Truck, Semi-Trailer, 5 Axles	0	6.76	0
Truck, Trailer, > 4 Axles	54,345	6.76	367,372
Truck, Trailer, > 5 Axles	0	6.76	0

Total CESAL in both direction 3.022 = 3,022,125

$W_{18} = D_D \times D_L \times W_{18}$ [2-Lane, 85%] = 2,568,806

Design CESAL, use 50% - 1 direction = 1,511,083

REPUBLIC OF THE PHILIPPINES DEPARTMENT OF PUBLIC WORKS AND HIGHWAYS REGION X CAGAYAN DE ORO CITY 1ST DISTRICT ENGINEERING OFFICE 10TH REGIONAL EQUIPMENT SERVICES COMPOUND BULUA, CAGAYAN DE ORO CITY	PROJECT NAME AND LOCATION: CONCRETING OF ROAD WITH DRAINAGE MANTICAO DRIVE PHASE II, BRGY. CANITOAN, CAGAYAN DE ORO CITY MISAMIS ORIENTAL	SHEET CONTENTS: PAVEMENT DESIGN ANALYSIS	SURVEYED: ENGINEER II DRAFTED / PREPARED:	REVIEWED: B. DAGALA DATE	SUBMITTED: S. DACANAY AND DESIGN SECTION DATE	RECOMMENDED: K. TIRADO DISTRICT ENGINEER DATE	APPROVED: VIR[illegible]IL T ENGINEER DATE	SET NO. : G 5 | 15	SHEET NO. : 6 25

GENERAL NOTES

1. DIMENSIONS:
   - A. DIMENSIONS ARE IN MILLIMETERS/METERS UNLESS OTHERWISE INDICATED.
   - B. STATIONS ARE GIVEN IN KILOMETERS AND METERS
   - C. RADII TRANSITION CURVES, AND ELEVATIONS ARE GIVEN IN METERS.
2. STATIONING:
   - A. ROAD STATIONS AND ELEMENTS ARE RELATIVE TO THE CENTERLINE OF THE ROAD UNLESS OTHERWISE INDICATED.
   - B. STATIONS ARE LOCATED AT A
     - A. EVERY 20 METERS
     - B. EVERY BRIDGE ABUTMENT AND
     - C. EVERY POINT OF HORIZONTAL GEOMETRY
3. ELEVATIONS AND GRADES:
   - A. ELEVATIONS GIVEN IN THE COLUMN "FINISHED GRADE" REFER TO THE FINISHED GRADE LEVEL AS SHOWN IN ROAD CROSS-SECTIONS.
   - B. THE ELEVATIONS GIVEN IN THE COLUMN "ORIGINAL GROUND" REFER TO THE EXISTING GROUND PROFILE AT THE ROAD GEOMETRIC CENTERLINE.
4. ROAD SIGNS:
   - A. THE MATERIALS, DIMENSIONS, SHAPES, COLORS, SIZE OF NUMERALS, LETTERS AND INSTALLATION SHALL BE IN ACCORDANCE WITH THE PHILIPPINE ROAD SIGN MANUAL, 1997.
5. PAVEMENT MARKINGS:
   - A. THE DIMENSIONS, SHAPES, COLORS, SIZE OF NUMERALS, LETTERS AND INSTALLATION SHALL BE IN ACCORDANCE WITH THE SPECIFICATIONS OF DPWH MANUAL ON PAVEMENT MARKINGS, 1980.
6. SPECIFICATIONS:
   - DPWH SPECIFICATION FOR HIGHWAYS, BRIDGES AND AIRPORTS, VOL. II
   - SPECIAL PROVISIONS
   - SUPPLEMENTAL SPECIFICATIONS
7. ALIGNMENT AND GRADES AREA SUBJECT TO ADJUSTMENT TO SUIT EXISTING FIELD CONDITIONS.
8. DISTANCES AND ELEVATIONS ARE IN METERS UNLESS INDICATED OTHERWISE.
9. GRADES SHOWN ARE TOP OF FINISHED PORTLAND CEMENT CONCRETE PAVEMENT
10. ALL WORKS SHALL COMPLY WITH THE "DPWH DESIGN GUIDELINES CRITERIA AND STANDARDS" VOLUME II - 2013 EDITION, "GENERAL SPECIFICATIONS FOR ROADS AND BRIDGES" 1976 AND "AASHTO INTERIM GUIDE FOR DESIGN OF PAVEMENT STRUCTURES" 1972.
11. BEFORE THE START OF ACTUAL CONSTRUCTION, THE AS-STAKED PLAN SHOULD BE SUBMITTED TO THE REGIONAL OFFICE IN ORDER THAT IMMEDIATE STEPS MAY BE UNDERTAKEN TO CORRECT OR ADJUST WHATEVER APPRECIABLE DEVIATION THERE MAY BE FROM THE ORIGINAL PLAN.
12. PRIOR TO CONSTRUCTION, THE ADEQUACY/APPROPRIATENESS OF THE THICKNESS OF PAVEMENT STRUCTURE SHALL BE VALIDATED BY UNDERTAKING PAVEMENT DESIGN ANALYSIS TAKING INTO CONSIDERATION THE ACTUAL CBR & TRAFFIC (ESAL) ALONG THE ROAD SECTION.
13. THESE PLAN ARE INTENDED FOR BIDDING AND CONTRACTION PURPOSES AND THE REQUIRED SURVEY DATA AND TECHNICAL INFORMATION IN ACCORDANCE WITH DPWH GUIDELINES WILL BE DETERMINED/VERIFIED DURING PRE-CONSTRUCTION SURVEY.
14. THE IMPLEMENTING OFFICE SHALL IDENTIFY THE LOCATIONS OF AND PROVIDE ACCESSIBILITY FACILITIES FOR PERSONS WITH DISABILITY IN ACCORDANCE WITH D.O. 37, SERIES OF 2009.

2013 AADT DATA :

TRAFFIC STATION :
ROAD SECTION ID :

MOTOR-TRICYCLE	=	
PASSENGER CAR	=	
PASSENGER UTILITY	=	
GOODS UTILITY	=	
SMALL BUS	=	
LARGE BUS	=	
RIGID TRUCKS 2 AXLES	=	
RIGID TRUCKS 3AXLES	=	
TRUCK SEMI-TRAILER 3&4 AXLES	=	
TRUCK SEMI-TRAILER 5 AXLES	=	
TRUCK TRAILER 4AXLES	=	
TRUCK TRAILER 5AXLES		

DESIGN DATA

DESIGN LIFE	=	20 YEARS
ASSUMED TRAFFIC GROTH, TGR	=	4.00%
R	=	85.00%
ZR	=	-1.037
OVERALL STANDARD DEVIATION, SO	=	0.35
W18	=	$\times 10^{6}$ (18 KIP ESAL)
pt	=	2.50
PRESENT SERVICEABILITY INDEX, ΔPSI	=	2.00
S'c	=	751 psi
Cd	=	1.00
J	=	3.80
Ec	=	3.605×10 PSI
k	=	pci
DESIGN CBR	=	
DESIGN CESAL	=	

SYMBOLS

PORTLAND CEMENT CONCRETE PAVEMENT	░
SECTION IN EARTH	▦
SECTION IN STONE MASONRY	▩
SECTION IN GRANULAR BEDDING	▒
LIMITS OF DIMENSION	|—|
CENTERLINE	℄
DIAMETER	Ø
PERCENT	%
AND	&
AT	@
NORTHING	

ABBREVIATIONS :

BEG.	BEGINNING
BH	BORE HOLE
CONC.	CONCRETE
ELEV.	ELEVATION
ENGR.	ENGINEER
F.G.E.	FINISHED GRADE ELEVATION
O.G.E.	ORIGINAL GROUND ELEVATION / TOP OF EXISTING PAVEMENT
H	HEIGHT
HOR.	HORIZONTAL
KM	KILOMETER
L	LENGTH
LVC	LENGTH OF VERTICAL CURVE
M / m.	METER
M.O.	MIDDLE ORDINATE
mm	MILLIMETER
PCCP	PORTLAND CEMENT CONCRETE PAVEMENT
PIVC	POINT OF INTERSECTION OF VERTICAL CURVE
RCBC	REINFORCED CONCRETE BOX CULVERT
RCPC	REINFORCED CONCRETE PIPE CULVERT
RSB	REINFORCING STEEL BAR
g	GRADE / SLOPE
e	SUPER ELEVATION
w	WIDENING
STA.	STATION
THK.	THICK
VERT.	VERTICAL
&	AND
A[illegible]	AREA OF PORTLAND CEMENT CONCRETE PAVEMENT
A_{CUT}	AREA CUT
A_{FILL}	AREA FILL

REPUBLIC OF THE PHILIPPINES
DEPARTMENT OF PUBLIC WORKS AND HIGHWAYS
REGION X
CAGAYAN DE ORO CITY 1ST DISTRICT
ENGINEERING OFFICE
10TH REGIONAL EQUIPMENT SERVICES COMPOUND
BULUA, CAGAYAN DE ORO CITY

PROJECT NAME AND LOCATION:
CONCRETING OF ROAD WITH DRAINAGE
MANTICAO DRIVE PHASE II, BRGY.
CANITOAN, CAGAYAN DE ORO CITY,
MISAMIS ORIENTAL

SHEET CONTENTS:
GENERAL NOTES

SURVEYED:
[illegible]
ENGINEER II

DRAFTED / [illegible]:
[illegible]

REVIEWED:
[illegible] DAGALA
DATE

SUBMITTED:
[illegible]
[illegible] AND DESIGN [illegible]
DATE

RECOMMENDED:
[illegible] TIRADO
DISTRICT ENGINEER
DATE

APPROVED:
[illegible]
DATE

SET NO.: G 6 15

SHEET NO.: 7 25

GEOMETRIC DESIGN STANDARD DRAWINGS

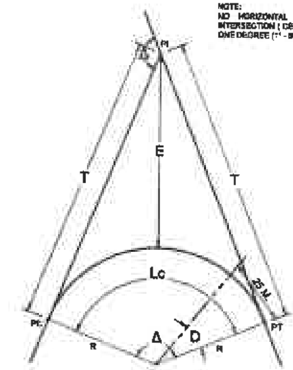
HORIZONTAL CURVE DETAILS

HORIZONTAL CURVE (CIRCULAR CURVE)

LEGEND:

PI = POINT OF INTERSECTION
Δ = INTERSECTION ANGLE (CENTRAL ANGLE)
T = TOTAL TANGENT DISTANCE
R = HORIZONTAL RADIUS
Lc = LENGTH OF CIRCULAR CURVE
E = TOTAL EXTERNAL DISTANCE
D = DEGREE OF CURVE (ARC DEFINITION)
PC = POINT OF CURVATURE
PT = POINT OF TANGENCY

NOTE:
NO HORIZONTAL CURVE IS REQUIRED WHEN THE INTERSECTION (CENTRAL ANGLE) ANGLE Δ IS LESS THAN ONE DEGREE (1° - 00' - 00")

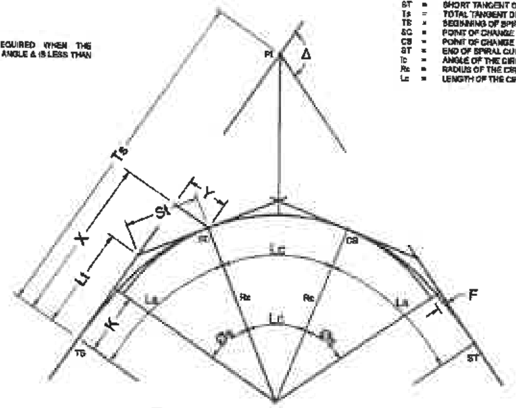


1A HORIZONTAL CURVE (CIRCULAR)
SCALE: NTS

HORIZONTAL CURVE WITH TRANSITION

LEGEND:

PI = POINT OF INTERSECTION
Δ = INTERSECTION ANGLE
θ = SPIRAL ANGLE
Ls = LENGTH OF SPIRAL
XY = COORDINATES OF CS OR SC WITH RESPECT TO MAIN TANGENT
K = LENGTHENING OF TANGENT DUE TO INTERSECTION OF SPIRAL
F = OFFSET BETWEEN CIRCULAR CURVE AND MAIN TANGENT
LT = LONG TANGENT OF SPIRAL
ST = SHORT TANGENT OF SPIRAL
Ts = TOTAL TANGENT DISTANCE
TS = BEGINNING OF SPIRAL CURVE
SC = POINT OF CHANGE FROM SPIRAL TO CIRCULAR CURVE
CS = POINT OF CHANGE FROM CIRCULAR CURVE TO SPIRAL
ST = END OF SPIRAL CURVE
Ic = ANGLE OF THE CIRCULAR CURVE
Rc = RADIUS OF THE CIRCULAR CURVE
Lc = LENGTH OF THE CIRCULAR CURVE



2A HORIZONTAL CURVE WITH TRANSITION
SCALE: NTS

VERTICAL PARABOLIC CURVE DETAILS

5A VERTICAL PARABOLIC CURVE
SCALE: NTS

IN ANY VERTICAL PARABOLIC CURVE:

SYMMETRICAL

$$MO = \frac{AL}{8.00}$$

$$Y_x = \frac{X^2 (MO)}{(L/2)^2}$$

UNSYMMETRICAL

$$MO = \frac{A L_1 L_2}{8.00}$$

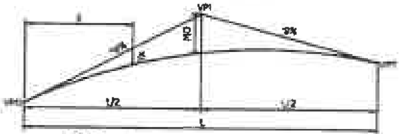
$$Y_x = \frac{X^2 (MO)}{(L_1)^2 \text{ OR } (L_2)^2}$$

LEGEND :

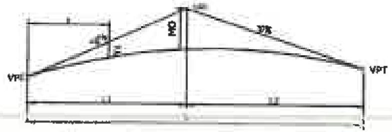
VPI = VERTICAL POINT OF INTERSECTION
VPC = VERTICAL POINT OF CURVATURE
VPT = VERTICAL POINT OF TANGENCY
L = LENGTH OF VERTICAL CURVES IN METERS
G = GRADE IN PERCENT
MO = MIDDLE ORDINATE
X = DISTANCE FROM VPC OR VPT TO ANY POINT OF CURVE
Yx = VERTICAL OFFSET IN METERS
A = ALGEBRAIC DIFFERENCE OF GRADES

NOTE:

NO VERTICAL CURVE IS REQUIRED WHEN THE ALGEBRAIC DIFFERENCE IN GRADES IS LESS THAN 0.50%

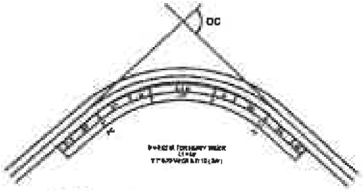


5A-1 SYMMETRICAL VERTICAL PARABOLIC CURVE
SCALE: NTS

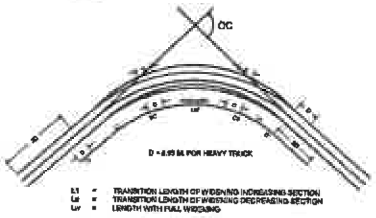


5A-2 UNSYMMETRICAL VERTICAL PARABOLIC CURVE
SCALE: NTS

WIDENING ON CURVE DETAILS

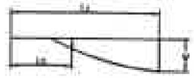


3A WIDENING WHEN TRANSITIONS NOT USED
SCALE: NTS



Lt = TRANSITION LENGTH OF WIDENING INCREASING SECTION
Lt = TRANSITION LENGTH OF WIDENING DECREASING SECTION
Lw = LENGTH WITH FULL WIDENING

4A WIDENING WHEN TRANSITIONS ARE USED
SCALE: NTS



f = KW (WIDENING AT ANY POINT)
SEE TABLE FOR "K - VALUES"

K - VALUES FOR WIDENING ON CURVES

INCREASING			DECREASING		
[illegible]	K		[illegible]	K	
0.00	0.0000	5.0	0.00	1.0000	18.0
0.02	0.0010	5.0	0.02	0.9964	29.5
0.04	0.0020	13.5	0.04	0.9905	47.5
0.06	0.0047	15.0	0.06	0.9810	79.0
0.08	0.0077	18.5	0.08	0.9690	111.0
0.10	0.0114	21.0	0.10	0.9458	119.0
0.12	0.0156	30.0	0.12	0.9200	140.0
0.14	0.0217	41.5	0.14	0.8920	159.0
0.16	0.0300	45.0	0.16	0.8602	192.0
0.18	0.3900	54.5	0.18	0.8238	121.0
0.20	0.0499	56.5	0.20	0.7816	246.0
0.22	0.0611	74.0	0.22	0.7374	251.0
0.24	0.0750	74.0	0.24	0.6822	242.0
0.26	0.0908	101.0	0.26	0.6340	246.0
0.28	0.1110	102.5	0.28	0.5848	241.5
0.30	0.1315	129.5	0.30	0.5355	225.5
0.32	0.1574	137.5	0.32	0.4912	212.0
0.34	0.1949	156.0	0.34	0.4478	193.0
0.36	0.2161	167.5	0.36	0.4092	172.0
0.38	0.2406	179.0	0.38	0.3748	152.6
0.40	0.2895	184.5	0.40	0.3443	149.6
0.42	0.3215	185.5	0.42	0.3144	138.0
0.44	0.3586	180.5	0.44	0.2888	129.0
0.46	0.3865	189.5	0.46	0.2610	118.5
0.48	0.4344	190.0	0.48	0.2379	106.0
0.50	0.4724	189.5	0.50	0.2169	98.0
0.52	0.5103	183.5	0.52	0.1967	91.5
0.54	0.5470	183.0	0.54	0.1784	85.5
0.56	0.5836	179.0	0.56	0.1613	80.0
0.58	0.6194	177.0	0.58	0.1452	74.5
0.60	0.6546	170.0	0.60	0.1304	71.0
0.62	0.6888	164.5	0.62	0.1162	64.0
0.64	0.7217	152.8	0.64	0.1034	59.0
0.66	0.7522	173.8	0.66	0.0915	54.5
0.68	0.7789	130.8	0.68	0.0807	49.5
0.70	0.8050	118.0	0.70	0.0708	42.0
0.72	0.8286	117.5	0.72	0.0622	38.5
0.74	0.8521	110.0	0.74	0.0543	35.0
0.76	0.8741	103.0	0.76	0.0473	33.0
0.78	0.8947	90.5	0.78	0.0471	28.5
0.80	0.9128	82.5	0.80	0.0348	30.0
0.82	0.9289	73.5	0.82	0.0288	26.0
0.84	0.9440	70.0	0.84	0.0236	23.0
0.86	0.9580	55.0	0.86	0.0190	20.0
0.88	0.9691	42.0	0.88	0.0150	17.0
0.90	0.9775	37.0	0.90	0.0116	17.0
0.92	0.9849	27.0	0.92	0.0182	15.0
0.94	0.9905	24.5	0.94	0.0052	12.0
0.96	0.9952	15.0	0.96	0.0028	8.0
0.98	0.9982	9.0	0.98	0.0012	6.0
1.00	1.00		1.00	0.0000	
	Lt			Lt	

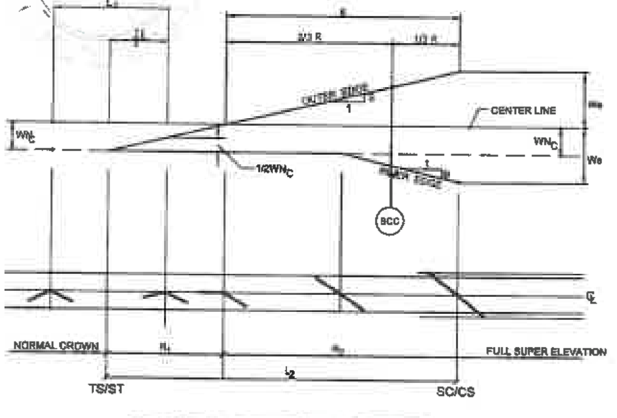
DESIGN VALUES FOR WIDENING ON CURVES

DEGREE OF CURVE	RADIUS IN METERS	DESIGN SPEED			
		30 KPH	40 KPH	50 KPH	60 KPH
1°00"	1432.390	0.00	0.00	0.00	0.00
2°00"	716.190	0.00	0.00	0.00	0.60
3°00"	477.460	0.00	0.00	0.60	0.60
4°00"	358.100	0.00	0.60	0.60	0.70
5°00"	286.480	0.60	0.60	0.70	0.80
6°00"	238.730	0.60	0.70	0.80	0.90
7°00"	204.630	0.70	0.80	0.90	1.10
8°00"	179.050	0.80	0.90	1.00	1.10
9°00"	159.150	0.90	0.90	1.00	1.10
10°00"	143.240	1.00	1.00	1.10	1.20
11°00"	130.220	1.10	1.10	1.20	1.30

NOTE: VALUES LESS THAN 0.60 M. MAY BE DISREGARDED VALUES SHOWN BELOW FOR 2 LANE PAVEMENT ON CURVES FOR A WIDTH [illegible]

REPUBLIC OF THE PHILIPPINES DEPARTMENT OF PUBLIC WORKS AND HIGHWAYS REGION X CAGAYAN DE ORO CITY 1ST DISTRICT ENGINEERING OFFICE 10TH REGIONAL EQUIPMENT SERVICES COMPOUND BULUA, CAGAYAN DE ORO CITY	PROJECT NAME AND LOCATION: CONCRETING OF ROAD WITH DRAINAGE MANTICAO DRIVE PHASE II, BRGY. CANITOAN, CAGAYAN DE ORO CITY, MISAMIS ORIENTAL	SHEET CONTENTS: GEOMETRIC DESIGN STANDARD DRAWINGS	SURVEYED: [illegible] ENGINEER II DRAFTED / PLOTTED: [illegible]	REVIEWED: [illegible] B. DAGALA ENGINEER II DATE	SUBMITTED: [illegible] MARCOS B. DACANAY [illegible] AND DESIGN SECTION DATE	RECOMMENDED: EVELYN [illegible] HIRADO ASSISTANT DISTRICT ENGINEER DATE	APPROVED: VIRGINIA [illegible] DATE	SET NO.: G 7 15	SHEET NO.: 8 25

GEOMETRIC DESIGN STANDARD DRAWINGS


L₁
2/3 R
1/3 R
OUTER EDGE
1
CENTER LINE
WNC
1/2WNC
WNC
We
We
INNER EDGE
BCC
NORMAL CROWN
R₁
R₂
FULL SUPER ELEVATION
L₂
TS/ST
SC/CS
℄

SUPERELEVATION OF SIMPLE CURVES

VALUES OF L AND S
(INTERPOLATED FROM AASHTO)

DESIGN SPEED km/h	L₁ m	S %
30	6	0.75
40	8	0.70
50	10	0.65
60	12	0.60
70	14	0.55
80	16	0.50
90	18	0.47
100	20	0.43
110	22	0.40
120	26	0.38

NOTE:
PER AASHTO POLICY L1 AT BEGIN AND END ARE NOT CONSIDERED

$R_1 = \frac{WN_C}{S}$

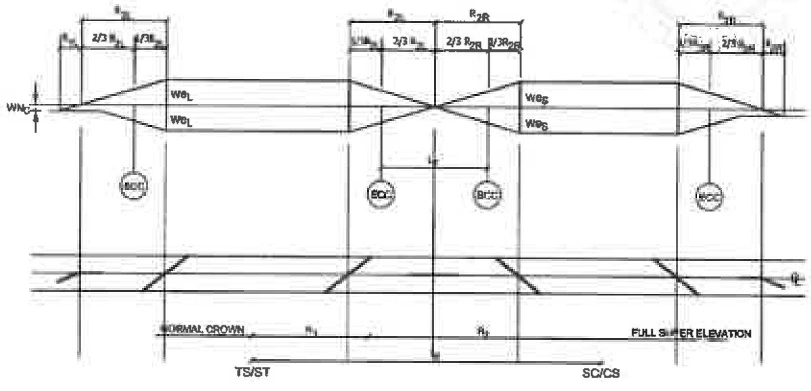
$R_2 = \frac{We}{S}$

$L_1 = \frac{WN_C}{S}$

$L_2 = R_1 + R_2 = \frac{W}{S}(N_C + e)$

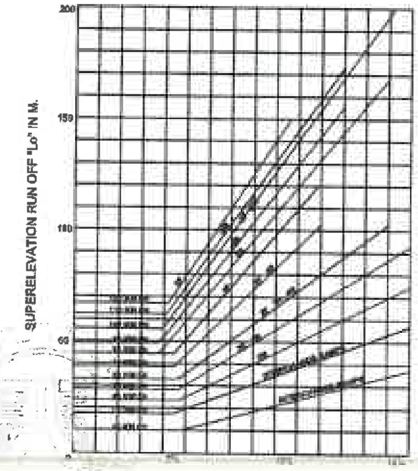
$L_x = L_1 + L_2 = \frac{W}{S}(2N_C + e)$

WHERE:
L_x = LENGTH OF CLOTHOID
L_2 = LENGTH OF CLOTHOID (per AASHO)
W = WIDTH OF LANE
e = SUPERELEVATION RATE

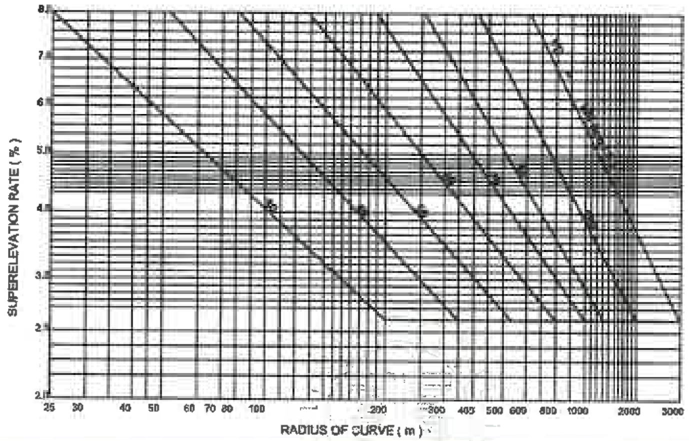

R₁L
R₂R
R₂L
R₁R
2/3 R₁L
1/3R₁L
1/3R₂L
2/3 R₂L
2/3 R₂R
1/3R₂R
WNC
WeL
WeL
WeR
WeR
BCC
ECC
BCC
ECC
L
NORMAL CROWN
R₁
R₂
FULL SUPER ELEVATION
TS/ST
SC/CS
℄

SUPERELEVATION OF REVERSE CURVES

NOTE:
MINIMUM L = 2/3 (R_{2L} + R_{2R})
DOTTED LINE FOR RIGHT EDGE



1A **SUPERELEVATION RUN OFF CHART (Lo)**
SCALE : NTS



2A **DESIGN SUPERELEVATION RATES**
SCALE : NTS

LEGEND :

ΔS = SLOPE OF EDGE OF PAVEMENT IN %
Lo = SUPERELEVATION RUN OFF
ef = FULL SUPERELEVATION
L₁ L₂ = LENGTH BETWEEN SUPERELEVATION 0 % AND 2 % WITH ΔS = 0.30 %
NC = NORMAL CROSSFALL

NOTES :

1. FOR EFFECTIVE DRAINAGE Δs HAS TO BE ≥ 0.30 % BETWEEN +/- 3 %.
2. WHERE Δs < 0.30 % , Δ SPECIAL METHOD OF SUPERELEVATION TRANSITION HAS TO BE ADOPTED AS INDICATED IN DOTTED LINE.
3. ROUNDING OFF ONLY NECESSARY IF Δs ≥ 0.60 %.

RADII FOR ROUNDING

V_{design}	< 80 KPH	> 80 KPH
R	100.00 M.	200.00 M.

4. ef = e MAX. CAN BE TAKEN FROM CHART FOR SUPERELEVATIONS.
5. THE SLOPE OF SHOULDER SHALL ALWAYS FALL IN THE DIRECTION OF THE OUTSIDE EDGE OF THE TRAVELLED WAY.
6. WHEN THE SUPERELEVATION IS LARGER THAN 4% THEN THE SLOPE OF THE LOWER SHOULDER SHALL BE THE SAME AS FOR THE TRAVELLED WAY.
7. THE ALGEBRAIC SUM OF THE SLOPES OF THE TRAVELED WAY AND THE SHOULDER SHOULD ALWAYS BE EQUAL TO 7%

REPUBLIC OF THE PHILIPPINES DEPARTMENT OF PUBLIC WORKS AND HIGHWAYS REGION X CAGAYAN DE ORO CITY 1ST DISTRICT ENGINEERING OFFICE 10TH REGIONAL EQUIPMENT SERVICES COMPOUND BULUA, CAGAYAN DE ORO CITY	PROJECT NAME AND LOCATION: CONCRETING OF ROAD WITH DRAINAGE MANTICAO DRIVE PHASE II, BRGY. CANITOAN, CAGAYAN DE ORO CITY, MISAMIS ORIENTAL	SHEET CONTENTS: GEOMETRIC DESIGN STANDARD DRAWINGS	SURVEYED: [illegible] ENGINEER II DRAFTED/PREPARED: [illegible] [illegible]	REVIEWED: [illegible] S. DATALA ENGINEER II DATE	SUBMITTED: JO[illegible] S. DACA[illegible]Y [illegible] AND DESIGN [illegible] DATE	RECOMMENDED: [illegible] TIRADO [illegible] DISTRICT ENGINEER DATE	APPROVED: VIR[illegible] DISTRICT ENGINEER DATE	SET NO. : G 8 | 16 SHEET NO. : 9 25

CONSTRUCTION REQUIREMENTS

GENERAL NOTES:

1. ALL WORKS SHALL COMPLY WITH THE "DPWH DESIGN GUIDELINES CRITERIA AND STANDARDS" - VOLUME II - 2013 EDITION, "GENERAL SPECIFICATIONS FOR ROADS AND BRIDGES," 1976 AND "AASHTO INTERIM GUIDE FOR DESIGN OF PAVEMENT STRUCTURES", 1972.
2. ALIGNMENT AND GRADES ARE SUBJECT TO ADJUSTMENT TO SUIT EXISTING FIELD CONDITIONS.
3. ALL ELEVATIONS, DIMENSIONS AND DISTANCES ARE ALL IN METERS UNLESS OTHERWISE SPECIFIED.
4. REFERENCE DATUM PLANE = MEAN SEA LEVEL.
5. ALL CONCRETE MIXTURES SHALL BE CLASS "A" 20.68 MPa (3,000 PSI) FOR THE PILE CAP AND THE REINFORCED SECTION OF THE REVETMENT, EXCEPT FOR THE RUBBLE CONCRETE WHICH SHALL BE CLASS "B", 17.25 MPa (2,500 PSI).
6. CONCRETE PROTECTION FOR REINFORCING STEEL BARS SHALL BE AT LEAST 50 mm.
7. BOULDERS OF TWO HUNDRED FIFTY TO THREE HUNDRED MM. (250-300MM) IN DIAMETER SHALL BE USED FOR THE FILL BEHIND THE REVETMENT WITH THE BIGGER SIZE NEAR THE TOE OF THE FILL. ALL VOIDS BETWEEN BOULDERS SHALL BE FILLED WITH MIXED SAND & GRAVEL AND COMPACTION SHALL BE DONE BY WATERING.
8. PRE-CONSTRUCTION SURVEY SHALL BE JOINTLY CONDUCTED BY THE CONTRACTOR & REPRESENTATIVES FROM THE IMPLEMENTING OFFICE. THE CORRESPONDING "AS-STAKED" PLAN SHALL BE PREPARED INCORPORATING THE CHANGES AS A RESULT OF THE PRE-CONSTRUCTION SURVEY & SHALL BE SUBJECT FOR APPROVAL BY THE UNDERSECRETARY FOR TECHNICAL SERVICES.

STONE MASONRY PROCEDURE:

1. THE BED FOR STONE MASONRY SHALL BE EXCAVATED TO THE REQUIRED DEPTHS AND PROPERLY COMPACTED, TRIMMED AND SHAPED.
2. THE STONE MASONRY SHALL BE FOUNDED IN A TOE TRENCH DUG BELOW THE DEPTH OF SCOUR AS SHOWN ON PLANS OR AS ORDERED BY THE ENGINEER. THE TOE TRENCH SHALL BE FILLED WITH STONE OF THE SAME CLASS AS THAT SPECIFIED FOR THE STONE MASONRY.
3. STONES SHALL BE PLACED BY HAND OR INDIVIDUALLY BY MACHINES. THEY SHALL BE LAID WITH CLOSE, BROKEN JOINTS AND SHALL BE FIRMLY BEDDED INTO THE SLOPE AND AGAINST THE ADJOINING STONES. EACH STONE SHALL BE LAID WITH EACH ADJACENT STONE. THE STONE MASONRY SHALL BE THOROUGHLY RAMMED INTO PLACE AS CONSTRUCTION PROGRESSES AND THE FINISHED SURFACE SHALL PRESENT AN EVEN, TIGHT SURFACE, INTERSTICES BETWEEN STONES SHALL BE FILLED WITH SMALL BROKEN FRAGMENTS FIRMLY RAMMED INTO PLACE.
4. THE SPACES BETWEEN THE STONES SHALL THEN BE FILLED WITH CEMENT MORTAR AS SPECIFIED IN SUBSECTION 506 2.2, MORTAR (DPWH STANDARD SPECS.) SUFFICIENT MORTAR SHALL BE USED TO COMPLETELY FILL ALL VOIDS, EXCEPT THAT THE FACE SURFACE OF THE STONES SHALL BE LEFT EXPOSED.
5. GROUT SHALL BE PLACED FROM BOTTOM TO TOP OF THE SURFACE SWEPT WITH A STIFF BROOM. AFTER GROUTING IS COMPLETED, THE SURFACE SHALL BE CURED AS SPECIFIED IN ITEM 405, STRUCTURAL CONCRETE FOR A PERIOD F AT LEAST THREE DAYS.

GROUTED RIPRAP :

1. STONES FOR RIPRAP SHALL BE CLASS "A" WHICH CONSIST OF ROCK EITHER RECTANGULAR IN SECTION OR ROUND NATURAL STONES AND SHALL RANGE FROM MINIMUM OF 15kg TO MAXIMUM OF 25kg WITH AT LEAST 50% OF THE STONES WEIGHING MORE THAN 20kg.
2. MORTAR FOR GROUTED RIPRAP SHALL CONSIST OF SAND CEMENT AND WATER CONFORMING TO THE REQUIREMENTS UNDER ITEM 405, STRUCTURAL CONCRETE, MIXED IN THE PROPORTION OF ONE PART CEMENT TO THE PARTS SAND BY VOLUME AND SUFFICIENT WATER TO PRODUCE A THICK BUT FLUID MORTAR.
3. 250 - 300mm Ø STONES SHALL BE USED FOR THE FACING AND SHALL BE HAND LAID WITH THE LONGEST DIMENSIONS PERPENDICULAR TO THE SLOPES AND FIRMLY BEDDED INTO THE SLOPE AND AGAINST THE ADJOINING BOULDER. SPACES BETWEEN THE BOULDERS SHALL BE COMPLETELY FILLED WITH MORTAR. THE OUTSIDE SURFACE OF THE BOULDERS SHALL BE LEFT EXPOSED AND THE SURFACE OF THE MORTAR SHALL BE SWEPT WITH A STIFF BROOM

HYDRAULIC DATA :

DESIGN DISCHARGE, $Q_{design} = 16.74\ m^3/s$ DESIGN VELOCITY, $v_{design} = 2.54$ m/s

STRUCTURAL CONCRETE:

1. THIS ITEM SHALL CONSIST OF FURNISHING, BENDING, PLACING & FINISHING CONCRETE IN ALL STRUCTURE EXCEPT PAVEMENTS IN ACCORDANCE WITH THIS SPECIFICATIONS AND CONFORMING TO THE LINES, GRADES & DIMENSIONS SHOWN ON THE PLANS. CONCRETE SHALL CONSIST OF MIXTURE OF PORTLAND CEMENT, FINE AGGREGATES COURSE AGGREGATE ADMIXTURE WHEN SPECIFIED, AND WATER MIXED IN THE PROPORTIONS SPECIFIED OR APPROVED BY THE ENGINEER.
2. CLASS A - ALL SUPERSTRUCTURE AND HEAVILY REINFORCED SUBSTRUCTURE. THE IMPORTANT PARTS OF THE STRUCTURE INCLUDED ARE SLABS, BEAMS, GIRDERS, ARCH RIBS, BOX CULVERT,REINFORCED ABUTMENTS, RETAINING WALLS AND REINFORCED FOOTING.

REINFORCING STEEL, GRADE 40:

DESCRIPTION:

1. THIS ITEM SHALL CONSIST OF FURNISHING, BENDING, FABRICATING AND OF PLACING OF STEEL REINFORCEMENT OF THE TYPE, SIZE & GRADE REQUIRED IN ACCORDANCE WITH THIS SPECIFICATION AND CONFORMITY WITH THE REQUIREMENTS SHOWN IN THE PLANS OR AS DIRECTED BY THE ENGINEER.

MATERIALS REQUIREMENTS:

1. REINFORCING STEEL SHALL MEET THE REQUIREMENT OF ITEM 710
2. REINFORCING STEEL & WIRE ROPE.

FILTER CLOTH :

1. POLYESTER OR POLYPROPYLENE — 100%
2. MECHANICALLY HEAT BONDED
3. NON-WOVEN & COMPRISE OF CONTINUOUS FILAMENT
4. EFFECTIVE OPENING ZONE — 110 MICRONS (max)
5. THICKNESS UNDER PRESSURE 200 KN/SQ.M. — 0.60mm (min)
4. WEIGHT — 200g/sqm
5. MULTIDIRECTIONAL TENSILE STRENGTH — 13KN/m
6. CBR PUNCTURE STRENGTH

FOR MATTRESSES:

1. BODY WIRE DIAMETER : 2.7mm (min)
2. SELVEDGE WIRE DIAMETER : 3.40mm (min)
3. TENSILE STRENGTH : 413.70 - 586.10 MPa
4. WEIGHT OF ZINC COATING : 244 g/sq. m. (min) OF COATED WIRE
5. ELONGATION: 200mm GAGE-LENGTH : 12% (min)
6. TYING & CONNECTING WIRE DIAMETER : 2.20mm (min)

ABBREVIATIONS:

BEG.	BEGINNING
CHB.	CONCRETE HOLLOW BLOCKS
CON.	CONCRETE
ELEV.	ELEVATION
ENG.	ENGINEER
FGE	FINISH GRADE ELEVATION
OGE	ORIGINAL GROUND ELEVATION
H	HEIGHT
HOR.	HORIZONTAL
K	KILOMETER
L	LENGTH
LVC	LENGTH OF VERTICAL CURVE
BEG.	METER
MO	MIDDLE ORDINATE
MM.	MILLIMETER
PCCP	PORTLAND CEMENT CONCRETE PAVEMENT
PIVC	POINT OF INTERSECTION OF VERTICAL CURVE
RCBC	REINFORCEMENT CONCRETE BOX CULVERT
RSB	REINFORCEMENT STEEL BAR
S	SLOPE
THK.	THICK
VERT.	VERTICAL
&	AND

LEGEND:

PROJECT ROAD	═╪═
℄ OF PROJECT ROAD	—●—●—
SECTION IN EARTH	▨
LIMITS OF DIMENSION	├—┤
CENTERLINE	℄
DIAMETER	Ø
PERCENT	%
AND	&
AT	@
NORTHING	Ⓝ

REPUBLIC OF THE PHILIPPINES DEPARTMENT OF PUBLIC WORKS AND HIGHWAYS REGION X CAGAYAN DE ORO CITY 1ST DISTRICT ENGINEERING OFFICE 10TH REGIONAL EQUIPMENT SERVICES COMPOUND BULUA, CAGAYAN DE ORO CITY	PROJECT NAME AND LOCATION: CONCRETING OF ROAD WITH DRAINAGE MANTICAO DRIVE PHASE II, BRGY. CANITOAN, CAGAYAN DE ORO CITY, MISAMIS ORIENTAL	SHEET CONTENTS: CONSTRUCTION REQUIREMENTS	SURVEYED: [illegible] ENGINEER II DRAFTED: [illegible] GRACE R. [illegible]	REVIEWED: [illegible] B. DAGALA ENGINEER II DATE	SUBMITTED: [illegible] S. DACANAY [illegible] DESIGN [illegible]	RECOMMENDED: [illegible] A. TIRADO [illegible] DISTRICT ENGINEER DATE	APPROVED: VI[illegible]IL DISTRICT ENGINEER DATE	SET NO.: G 9 | 15 SHEET NO.: 10 25

CONSTRUCTION REQUIREMENTS

ITEM 100 - CLEARING AND GRUBBING

1 DESCRIPTION

THIS ITEM SHALL CONSIST OF CLEARING, GRUBBING, REMOVING AND DISPOSING ALL VEGETATION AND DEBRIS AS DESIGNATED IN THE CONTRACT, EXCEPT THOSE OBJECTS THAT ARE DESIGNATED TO REMAIN IN PLACE OR ARE TO BE REMOVED IN CONSONANCE WITH OTHER PROVISIONS OF THIS SPECIFICATION. THE WORK SHALL ALSO INCLUDE THE PRESERVATION FROM INJURY OR DEFACEMENT OF ALL OBJECTS DESIGNATED TO REMAIN.

2 CONSTRUCTION REQUIREMENTS

2.1 GENERAL

THE ENGINEER WILL ESTABLISH THE LIMITS OF WORK AND DESIGNATE ALL TREES, SHRUBS, PLANTS AND OTHER THINGS TO REMAIN. THE CONTRACTOR SHALL PRESERVE ALL OBJECTS DESIGNATED TO REMAIN. PAINT REQUIRED FOR CUT OR SCARRED SURFACE OF TREES OR SHRUBS SELECTED FOR RETENTION SHALL BE AN APPROVED ASPHALTUM BASE PAINT PREPARED ESPECIALLY FOR TREE SURGERY.

CLEARING SHALL EXTEND ONE (1) METER BEYOND THE TOE OF THE FILL SLOPES OR BEYOND ROUNDING OF CUT SLOPES AS THE CASE MAYBE FOR THE ENTIRE LENGTH OF THE PROJECT UNLESS OTHERWISE SHOWN ON THE PLANS OR AS DIRECTED BY THE ENGINEER AND PROVIDED IT IS WITHIN THE RIGHT OF WAY LIMITS OF THE PROJECT, WITH THE EXCEPTION OF TREES UNDER THE JURISDICTION OF THE FOREST MANAGEMENT BUREAU (FMB).

2.2 CLEARING AND GRUBBING

ALL SURFACE OBJECTS AND ALL TREES, STUMPS, ROOTS AND OTHER PROTRUDING OBSTRUCTIONS, NOT DESIGNATED TO REMAIN, SHALL BE CLEARED AND/OR GRUBBED, INCLUDING MOWING AS REQUIRED, EXCEPT AS PROVIDED BELOW:

A. REMOVAL OF UNDISTURBED STUMPS AND ROOTS AND NONPERISHABLE SOLID OBJECTS WITH A MINIMUM DEPTH OF ONE (1) METER BELOW SUBGRADE OR SLOPE OF EMBANKMENT WILL NOT BE REQUIRED.

B. IN AREAS OUTSIDE OF THE GRADING LIMITS OF CUT AND EMBANKMENT AREAS, STUMPS AND NONPERISHABLE SOLID OBJECTS SHALL BE CUT OFF NOT MORE THAN 150 MM (6 INCHES) ABOVE THE GROUND LINE OR LOW WATER LEVEL.

C. IN AREAS TO BE ROUNDED AT THE TOP OF CUT SLOPES, STUMPS SHALL BE CUT OFF FLUSH WITH OR BELOW THE SURFACE OF THE FINAL SLOPE LINE.

D. GRUBBING OF PITS, CHANNEL CHANGES AND DITCHES WILL BE REQUIRED ONLY TO THE DEPTH NECESSITATED BY THE PROPOSED EXCAVATION WITHIN SUCH AREAS.

E. IN AREAS COVERED BY COGON/TALAHIB, WILD GRASS AND OTHER VEGETATIONS, TOP SOIL SHALL BE CUT TO A MAXIMUM DEPTH OF 150 MM BELOW THE ORIGINAL GROUND SURFACE OR AS DESIGNATED BY THE ENGINEER, AND DISPOSED OUTSIDE THE CLEARING AND GRUBBING LIMITS AS INDICATED IN THE TYPICAL ROADWAY SECTION.

2.3 INDIVIDUAL REMOVAL OF TREES OR STUMPS

INDIVIDUAL TREES OR STUMPS DESIGNATED BY THE ENGINEER FOR REMOVAL AND LOCATED IN AREAS OTHER THAN THOSE ESTABLISHED FOR CLEARING AND GRUBBING AND ROADSIDE CLEANUP SHALL BE REMOVED AND DISPOSED OFF AS SPECIFIED UNDER SUBSECTION 100.2.2 EXCEPT TREES REMOVED SHALL BE CUT AS NEARLY FLUSH WITH THE GROUND AS PRACTICABLE WITHOUT REMOVING STUMPS.

ITEM 102 - EXCAVATION

1. ROADWAY EXCAVATION

ROADWAY EXCAVATION WILL INCLUDE EXCAVATIONS OF THE FOLLOWING:

A. GRADING FOR ROADWAYS
B. APPROACHES
C. SLOPE ROUNDING
D. BENCHING
E. WATERWAYS AND DITCHES
F. AND REMOVAL OF UNSUITABLE MATERIALS FROM THE ROADBED AND BENEATH EMBANKMENT AREAS

COMMON EXCAVATION CONSIST OF ALL EXCAVATION NOT INCLUDED IN THE BILL OF QUANTITIES UNDER "ROCK EXCAVATION" OR OTHER PAY ITEMS.

BORROW EXCAVATION SHALL CONSIST OF THE EXCAVATION AND UTILIZATION OF APPROVED MATERIAL REQUIRED FOR THE CONSTRUCTION OF EMBANKMENTS OR FOR OTHER PORTIONS OF THE WORK, AND SHALL BE OBTAINED FROM APPROVED SOURCES.

2. UTILIZATION OF EXCAVATED MATERIALS

ALL SUITABLE MATERIAL REMOVED FROM THE EXCAVATION SHALL BE USED IN THE FORMATION OF THE EMBANKMENT, SUBGRADE, SHOULDERS, SLOPES, BEDDING, AND BACKFILL FOR STRUCTURES, AND FOR OTHER PURPOSES SHOWN ON THE PLANS OR AS DIRECTED.

3. SURPLUS COMMON EXCAVATION

- DISPOSAL OF EXCAVATED MATERIALS NOT SUITABLE FOR EMBANKMENT AREAS.
- DISPOSAL OF EXCAVATED MATERIALS ON DESIGNATED AREA AS DIRECTED BY THE ENGINEER.

ITEM 103 - STRUCTURE EXCAVATION

1. EXCAVATION

1.1 THE CONTRACTOR SHALL NOTIFY THE ENGINEER SUFFICIENTLY IN ADVANCE OF THE BEGINNING OF ANY EXCAVATION SO THAT CROSS-SECTIONAL ELEVATIONS AND MEASUREMENTS MAY BE TAKEN ON THE UNDISTURBED GROUND. THE NATURAL GROUND ADJACENT TO THE STRUCTURE SHALL NOT BE DISTURBED WITHOUT PERMISSION OF THE ENGINEER.

- PIPE CULVERTS: THE WIDTH OF THE PIPE TRENCH SHALL BE SUFFICIENT TO PERMIT SATISFACTORY JOINTING OF THE PIPE AND THOROUGH TAMPING OF THE BEDDING MATERIAL UNDER AND AROUND THE PIPE.
- WIDTH OF EXCAVATION SHALL BE AT LEAST 300 MM (12 INCHES) GREATER THAN THE HORIZONTAL OUTSIDE DIAMETER OF THE PIPE.
- THE EXCAVATION BELOW GRADE SHALL BE BACKFILLED WITH SELECTED FINE COMPRESSIBLE MATERIAL, E.G. SILTY CLAY OR LOAM, AND LIGHTLY COMPACTED IN LAYERS NOT OVER 150 MM (6 INCHES).

1.2 UTILIZATION OF EXCAVATED MATERIALS

- ALL EXCAVATED MATERIALS, SO FAR AS SUITABLE, SHALL BE UTILIZED AS BACKFILL OR EMBANKMENT.
- SURPLUS MATERIALS SHALL BE DISPOSED OFF AS DESIGNATED BY THE ENGINEER
- NO EXCAVATED MATERIALS SHALL BE DEPOSITED AT ANY TIME TO ENDANGER THE PARTLY FINISHED STRUCTURE.

1.3 BACKFILL AND EMBANKMENT FOR STRUCTURES OTHER THAN PIPE CULVERTS

- EMBANKMENT AND BACKFILLING SHALL BE LAYER BY LAYER NOT OVER 150 MM (6 INCHES) IN THICKNESS UP TO THE LEVEL OF ORIGINAL GROUND.

ITEM 104 - EMBANKMENT

1. MATERIAL REQUIREMENTS

EMBANKMENTS SHALL BE CONSTRUCTED OF SUITABLE MATERIALS, IN CONSONANCE WITH THE FOLLOWING DEFINITIONS:

1.1 SUITABLE MATERIAL - MATERIAL WHICH IS ACCEPTABLE IN ACCORDANCE WITH THE CONTRACT AND WHICH CAN BE COMPACTED IN THE MANNER SPECIFIED IN THIS ITEM. IT CAN BE COMMON MATERIAL OR ROCK.

2. CONSTRUCTION REQUIREMENTS

EMBANKMENT CONSTRUCTION SHALL CONSIST OF CONSTRUCTING ROADWAY EMBANKMENTS.

- THE PLACING AND COMPACTING OF APPROVED MATERIAL WITHIN THE ROADWAY AREAS WHERE UNSUITABLE MATERIALS HAS BEEN REMOVED.
- THE PLACING AND COMPACTING OF EMBANKMENT MATERIALS IN HOLES, PITS, AND OTHER DEPRESSIONS WITHIN THE ROADWAY AREA.
- THE EXISTING GROUND SHALL BE COMPACTED TO A DEPTH OF 150 MM (6 INCHES) AND TO THE SPECIFIED REQUIREMENTS OF THIS ITEM.

	PROJECT NAME AND LOCATION:	SHEET CONTENTS:	SURVEYED:	REVIEWED:	SUBMITTED:	RECOMMENDED:	APPROVED:	SET NO.:	SHEET NO.:
REPUBLIC OF THE PHILIPPINES DEPARTMENT OF PUBLIC WORKS AND HIGHWAYS REGION X CAGAYAN DE ORO CITY 1ST DISTRICT ENGINEERING OFFICE 10TH REGIONAL EQUIPMENT SERVICES COMPOUND BULUA, CAGAYAN DE ORO CITY	CONCRETING OF ROAD WITH DRAINAGE MANTICAO DRIVE PHASE II, BRGY. CANITOAN, CAGAYAN DE ORO CITY, MISAMIS ORIENTAL	CONSTRUCTION REQUIREMENTS	[illegible] ENGINEER I PLOTTED/PREPARED: [illegible] DATE	[illegible] GAGALA ENGINEER II DATE	[illegible] DACANAY [illegible] AND DESIGN SECTION DATE	EVELYN A. TIRADO ASSISTANT DISTRICT ENGINEER DATE	VIRGINIA [illegible] [illegible] ENGINEER DATE	G 18 | 19	11 25

CONSTRUCTION REQUIREMENTS

- EXISTING OLD ROAD SURFACE SHALL BE SCARIFIED TO A DEPTH OF AT LEAST 150 MM (6 INCHES) BEFORE EMBANKMENT TAKES PLACE. THE SCARIFIED MATERIALS THEN BE COMPACTED AS PROVIDED IN DPWH STANDARD SPECIFICATIONS.

ITEM 105 - SUBGRADE PREPARATION

1. SUBGRADE IN COMMON EXCAVATION

UNLESS OTHERWISE SPECIFIED, ALL MATERIALS BELOW SUBGRADE LEVEL IN EARTH CUTS TO A DEPTH 150 MM OR OTHER DEPTH SHOWN ON THE PLANS OR AS DIRECTED BY THE ENGINEER SHALL BE EXCAVATED.

2. SUBGRADE IN ROCK EXCAVATION

SURFACE IRREGULARITIES UNDER THE SUBGRADE LEVEL REMAINING AFTER TRIMMING OF THE ROCK EXCAVATION SHALL BE LEVELED BY PLACING SPECIFIED MATERIAL AND COMPACTED TO THE REQUIREMENTS OF SUBSECTION 104.3.3.

3. SUBGRADE ON EMBANKMENT

IT SHALL BE IN ACCORDANCE WITH SUBSECTION 105.3.5 SUBGRADE ON EMBANKMENT.

4. SUBGRADE ON EXISTING PAVEMENT

IT SHALL BE IN ACCORDANCE WITH SUBSECTION 105.3.6 SUBGRADE ON EXISTING PAVEMENT.

ITEM 200 - AGGREGATE SUBBASE COURSE

1. MATERIAL REQUIREMENTS

AGGREGATE FOR SUBBASE SHALL CONSIST OF HARD, DURABLE PARTICLES OR FRAGMENTS OF CRUSHED STONE, CRUSHED SLAG, OR CRUSHED OR NATURAL GRAVEL AND FILLER OF NATURAL OR CRUSHED SAND OR OTHER FINELY DIVIDED MINERAL MATTER. THE COMPOSITE MATERIAL SHALL BE FREE FROM VEGETABLE MATTER AND LUMPS OR BALLS OF CLAY, AND SHALL BE OF SUCH NATURE THAT IT CAN BE COMPACTED READILY TO FORM A FIRM, STABLE SUBBASE.

2. PREPARATION OF EXISTING SURFACE

THE EXISTING SURFACE SHALL BE GRADED AND FINISHED AS PROVIDED UNDER ITEM 105, SUBGRADE PREPARATION, BEFORE PLACING THE SUBBASE MATERIAL.

3. PLACING

IT SHALL BE IN ACCORDANCE WITH ALL THE REQUIREMENTS OF SUBSECTION 201.3.2 PLAC ING.

4. SPREADING AND COMPACTING

IT SHALL BE IN ACCORDANCE WITH ALL THE REQUIREMENTS OF SUBSECTION 201.3.3, SPREDING AND COMPACTING.

5. PROTECTION OF COMPLETED WORK

THE CONTRACTOR SHALL BE REQUIRED TO PROTECT AND MAINTAIN AT HIS OWN EXPENSE OF THE ENTIRE WORK WITHIN THE LIMITS OF HIS CONTRACT IN GOOD CONDITION SATISFACTORY TO THE ENGINEER FROM TIME TO TIME UNTIL ALL WORK SHALL BE COMPLETED.

ITEM 311 - PORTLAND CEMENT CONCRETE PAVEMENT

MATERIALS

1.1 PORTLAND CEMENT
- USE TYPE 1 PORTLAND CEMENT, 40 KGS; "IF CEMENT IS USED IN BULK, THE ENGINEER SHALL BE INFORMED."
- MIXING PROPORTION: USE CLASS "A", 364 KGS/ CU.M. OF CONCRETE OR 9.10 BAGS/ CU.M.

1.2 FINE AGGREGATE
- REFER TO TABLE 311.1 - GRADING REQUIREMENTS IN DPWH STANDARD SPECIFICATION 2004 EDITION.

1.3 COARSE AGGREGATE
- REFER TO TABLE 311.2 - GRADING REQUIREMENTS IN DPWH STANDARD SPECIFICATION 2004 EDITION.

1.4 WATER
- IF DRINKABLE WATER IS USED, NO TEST IS REQUIRED.

ITEM 404 - REINFORCING STEEL BARS

1 DESCRIPTION

THIS ITEM SHALL CONSIST OF FURNISHING, BENDING, FABRICATING AND PLACING OF STEEL REINFORCEMENT OF THE TYPE, SIZE, SHAPE AND GRADE REQUIRED IN ACCORDANCE WITH THIS SPECIFICATION AND IN CONFORMITY WITH THE REQUIREMENTS SHOWN ON THE PLANS OR AS DIRECTED BY THE ENGINEER.

2 MATERIAL REQUIREMENTS

REINFORCING STEEL SHALL MEET THE REQUIREMENTS OF ITEM 710, REINFORCING STEEL AND WIRE ROPE.

ITEM 405 - STRUCTURAL CONCRETE

THIS ITEM SHALL CONSIST OF FURNISHING, BENDING, PLACING AND FINISHING CONCRETE IN ALL STRUCTURES EXCEPT PAVEMENTS IN ACCORDANCE WITH THIS SPECIFICATION AND CONFORMING TO THE LINES, GRADES, AND DIMENSIONS SHOWN ON THE PLANS. CONCRETE SHALL CONSIST OF A MIXTURE OF PORTLAND CEMENT, FINE AGGREGATE, COARSE AGGREGATE, ADMIXTURE WHEN SPECIFIED, AND WATER MIXED IN THE PROPORTIONS SPECIFIED OR APPROVED BY THE ENGINEER.

THE CLASSES OF CONCRETE WILL GENERALLY BE USED AS FOLLOWS:

CLASS A - ALL SUPERSTRUCTURES AND HEAVILY REINFORCED SUBSTRUCTURES. THE IMPORTANT PARTS OF THE STRUCTURE INCLUDED ARE SLABS, BEAMS, GIRDERS, COLUMNS, ARCH RIBS, BOX CULVERTS, REINFORCED ABUTMENTS, RETAINING WALLS, AND REINFORCED FOOTINGS.
CLASS B - FOOTINGS, PEDESTALS, MASSIVE PIER SHAFTS, PIPE BEDDING, AND GRAVITY WALLS, UNREINFORCED OR WITH ONLY A SMALL AMOUNT OF REINFORCEMENT.
CLASS C - THIN REINFORCED SECTIONS, RAILINGS, PRECAST R.C. PILES AND CRIBBING AND FOR FILLER IN STEEL GRID FLOORS.
CLASS P - PRESTRESSED CONCRETE STRUCTURES AND MEMBERS.
SEAL - CONCRETE DEPOSITED IN WATER

MATERIAL REQUIREMENTS

PORTLAND CEMENT
IT SHALL CONFORM TO ALL THE REQUIREMENTS OF SUBSECTION 311.2.1.

FINE AGGREGATE
IT SHALL CONFORM TO ALL THE REQUIREMENTS OF SUBSECTION 311.2.2.

COARSE AGGREGATE
IT SHALL CONFORM ALL THE REQUIREMENTS OF SUBSECTION 311.2.3 EXCEPT THAT GRADATION SHALL CONFORM TO TABLE 405.1

ITEM 506 - STONE MASONRY

1. STONE

THE STONE SHALL BE CLEAN, HARD, AND DURABLE AND SHALL BE SUBJECT TO THE ENGINEER'S APPROVAL. ADOBE STONE SHALL NOT BE USED UNLESS OTHERWISE SPECIFIED.

1.1 SIZES AND SHAPES
- THICKNESS: NOT LESS THAN 150 MM
- WIDTH: NOT LESS THAN 1-1/2 TIMES THEIR RESPECTIVE THICKNESS
- LENGTH: NOT LESS THAN 1-1/2 TIMES THEIR RESPECTIVE WIDTHS

NOTE: EACH STONE SHALL BE OF GOOD SHAPE AND BE FREE OF DEPRESSIONS AND PROTECTIONS THAT MIGHT WEAKENED OR PREVENT IT FROM BEING PROPERLY BEDDED.

1.2 DRESSING - THE STONE SHALL BE PROPERLY DRESSED TO REMOVE ANY THIN OR WEAK PORTION.
1.3 FINISHED FOR EXPOSED FACES - FACE STONES SHALL BE PITCHED TO THE LINE ALONG THE BEDS AND JOINTS. THE MAXIMUM PROJECTIONS OF ROCK FACES BEYOND THE PITCH SHALL NOT BE MORE THAN 50 MM.

2. MORTAR

MORTAR FOR THE MASONRY SHALL BE COMPOSED OF ONE (1) PART PORTLAND CEMENT AND TWO (2) PARTS FINE AGGREGATES BY VOLUME.

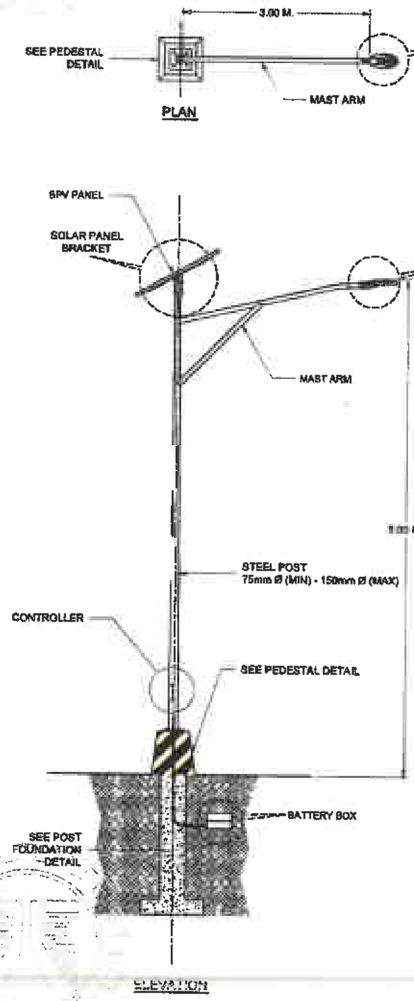
3. WEEP HOLES

IT SHALL CONFORM TO THE REQUIREMENT OF ITEM 504, RIPRAP AND GROUTED RIPRAP.

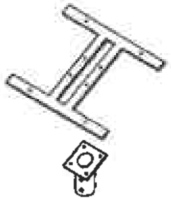
REPUBLIC OF THE PHILIPPINES DEPARTMENT OF PUBLIC WORKS AND HIGHWAYS REGION X CAGAYAN DE ORO CITY 1ST DISTRICT ENGINEERING OFFICE 10TH REGIONAL EQUIPMENT SERVICES COMPOUND BULUA, CAGAYAN DE ORO CITY	PROJECT NAME AND LOCATION: CONCRETING OF ROAD WITH DRAINAGE MANTICAO DRIVE PHASE II, BRGY. CANITOAN, CAGAYAN DE ORO CITY, MISAMIS ORIENTAL	SHEET CONTENTS: CONSTRUCTION REQUIREMENTS	SURVEYED: [illegible]ALD M. [illegible] ENGINEER I DRAFTED: [illegible] GRACE P. [illegible]	REVIEWED: [illegible] B. DAGALA ENGINEER II DATE	SUBMITTED: [illegible] S[illegible]Y [illegible] AND DESIGN SECTION DATE	RECOMMENDED: EV[illegible]N [illegible] TIRADO ASSISTANT DISTRICT ENGINEER DATE	APPROVED: VIR[illegible]AIL [illegible] DATE	SET NO.: G 11 | 15	SHEET NO.: 12 25

ROADWAY STREET LIGHTING

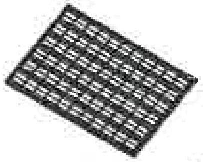
FOR FUTURE FUNDING


3.00 M.
SEE PEDESTAL DETAIL
LED LIGHT
MAST ARM
PLAN
SPV PANEL
SOLAR PANEL BRACKET
LED LIGHT
MAST ARM
9.00 M.
STEEL POST
75mm Ø (MIN) - 150mm Ø (MAX)
CONTROLLER
SEE PEDESTAL DETAIL
BATTERY BOX
SEE POST FOUNDATION DETAIL
ELEVATION

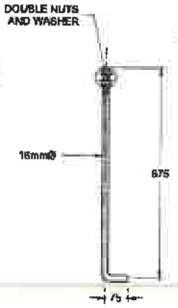
STREET LIGHT TYPICAL DETAIL
SCALE: NTS



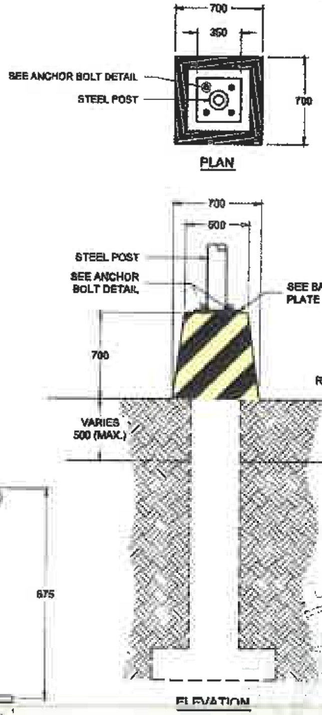
TYPICAL SOLAR PANEL BRACKET
SCALE: NTS



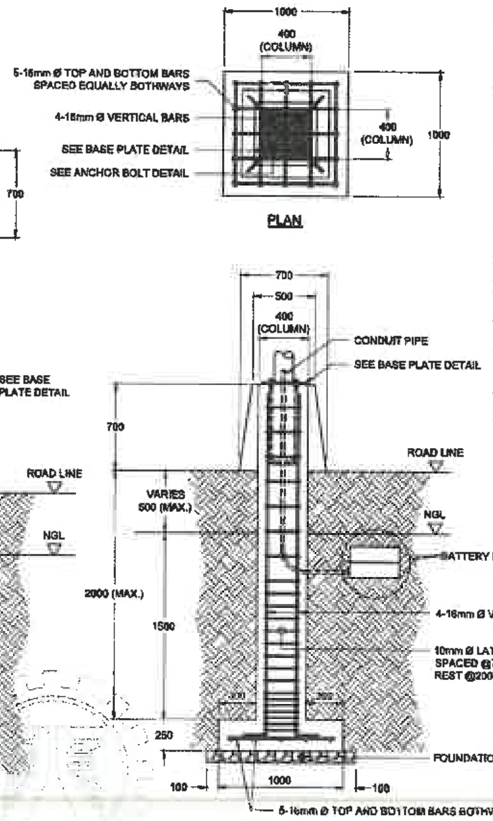
TYPICAL SOLAR PHOTO VOLTAIC (SPV) PANEL
SCALE: NTS


DOUBLE NUTS AND WASHER
16mmØ
675
75

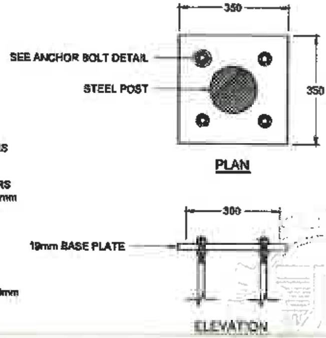
ANCHOR BOLT DETAIL
SCALE: NTS


700
350
SEE ANCHOR BOLT DETAIL
STEEL POST
700
PLAN
700
500
STEEL POST
SEE ANCHOR BOLT DETAIL
SEE BASE PLATE DETAIL
700
ROAD LINE
VARIES 500 (MAX.)
NGL
ELEVATION

PEDESTAL DETAIL
SCALE: NTS


1000
400 (COLUMN)
5-16mm Ø TOP AND BOTTOM BARS SPACED EQUALLY BOTHWAYS
4-16mm Ø VERTICAL BARS
SEE BASE PLATE DETAIL
SEE ANCHOR BOLT DETAIL
400 (COLUMN)
1000
PLAN
700
500
400 (COLUMN)
CONDUIT PIPE
SEE BASE PLATE DETAIL
700
ROAD LINE
VARIES 500 (MAX.)
NGL
BATTERY BOX
2000 (MAX.)
1500
4-16mm Ø VERTICAL BARS
10mm Ø LATERAL TIE BARS SPACED @7-50mm, 8-100mm REST @200mm O.C.
200
200
250
FOUNDATION FILL, t = 150mm
100
1000
100
5-16mm Ø TOP AND BOTTOM BARS BOTHWAYS
ELEVATION

POST FOUNDATION DETAIL
SCALE: NTS


350
SEE ANCHOR BOLT DETAIL
STEEL POST
350
PLAN
300
19mm BASE PLATE
ELEVATION

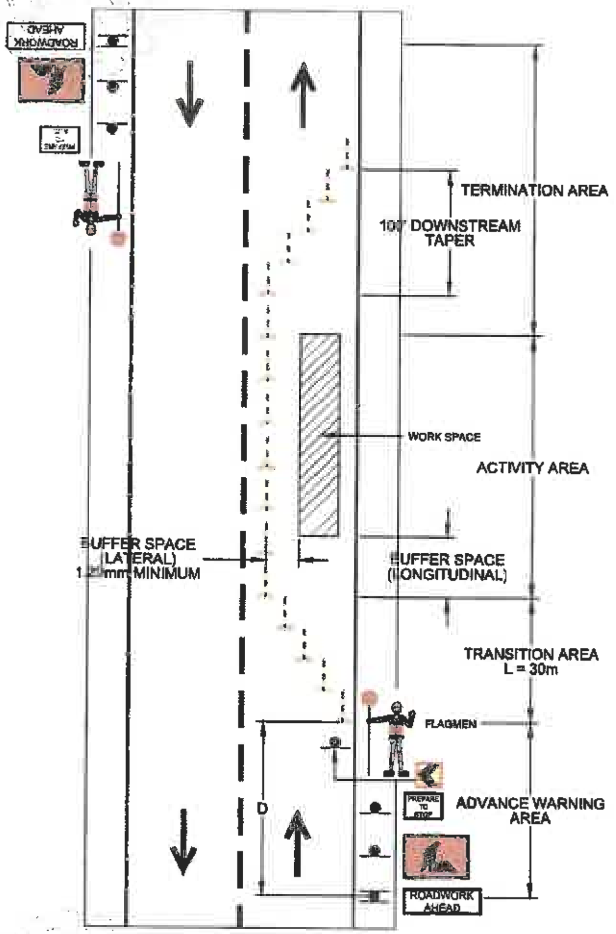
BASE PLATE DETAIL
SCALE: NTS

NOTES :

1. ALL DIMENSIONS ARE IN MILLIMETERS UNLESS OTHERWISE SPECIFIED.
2. THE LOCATION OF STREET LIGHTING POSTS SHOWN ON THE DRAWINGS ARE APPROXIMATE AND THE EXACT LOCATION SHALL BE DETERMINED/ESTABLISHED BY THE ENGINEER IN THE FIELD.
3. ALL MATERIALS SHALL BE BRAND NEW AND SHALL BE OF THE APPROVED TYPE. IT SHALL CONFORM WIHT THE REQUIREMENTS OF THE PHILIPPINE ELECTRICAL CODE AND THE PRODUCTS LOCALLY MANUFACTURED SHALL BEAR A PHILIPPINE STANDARD (PS) MARK, WHILE IMPORTED PRODUCTS SHALL BEAR THE IMPORT COMMODITY CLEARANCE (ICC) CERTIFICATION MARKS DULY ISSUED BY THE BUREAU OF PHILIPPINE STANDARDS (BPS).
4. FERROUS METAL RACEWAYS, CABEL ARMOR, BOXES, CABLE SHEATHING, CABINETS, ELBOWS, COUPLINGS, NIPPLES, FITTINGS, SUPPORTS, AND SUPPORT HARDWARE SHALL BE PERMITTED TO BE INSTALLED IN CONCRETE OR IN DIRECT CONTACT WITH SOIL CORROSION PROTECTION SHALL BE PROVIDED IN AREAS SUBJECT TO SEVERE CORROSIVE INFLUENCES WHERE MADE OF THE MATERIAL APPROVED FOR THE CONDITION, OR WHERE PROVIDED WITH CORROSION PROTECTION APPROVED FOR THE CONDITION AS INDICATED IN THE PLANS OR AS APPROVED BY THE ENGINEER.
5. STREET LIGHTING POSTS SHALL ADHERE TO THE DETAILS AS INDICATED ON THE PLANS. ALL POSTS SHALL BE DESIGNED TO WITHSTAND, WITH ASSOCIATED LUMINAIRES, A WIND VELOCITY OF 250 KPH DETERMINED IN ACCORDANCE WITH AASHTO LTS-6, STANDARD SPECIFICATIONS FOR STRUCTURAL SUPPORTS FOR HIGHWAY SIGNS, LUMINAIRES, AND TRAFFIC SIGNALS, WHILE SUPPORTING LUMINAIRES HAVING EFFECTIVE PROJECTED AREA INDICATED. POLES SHALL BE ANCHOR-BASED TYPES DESIGNED FOR USE WITH UNDERGROUND SUPPLY CONDUCTORS.
6. ALL POSTS SHALL HAVE AN INSPECTION DOOR OR HAND HOLE AND SHALL HAVE A SUITABLE GASKETED SCREW COVER. AFTER THE POST HAS BEEN ERECTED, SEALED AND FIXED IN THE FOUNDATION BLOCK, A COAT OF FINISHING ALUMINUM PAINT SHALL BE APPLIED. POSTS SHALL BE PROVIDED WITH GALVANIZED STEEL ANCHOR BOLTS THREADED TOP END AND WITJ A HOOKED BENT AT AN ANGLE END AT BOTTOM. SIZE OF THE ANCHOR BOLTS SHALL BE INDICATED IN THE PLANS.
7. POLE SHAFT SHALL BE MADE IN ACCORDANCE TO ASTM A53M, STANDARD SPECIFICATION FOR PIPE, STEEL, BLACK AND HOT-DIPPED, ZINC COATED, WELDED AND SEAMLESS.
8. GALVANIZING SHALL BE IN ACCORDANCE WITH THE REQUIREMENTS OF ASTM A153, STANDARD SPECIFICATION FOR ZINC COATING (HOT-DIP) ON IRON AND STEEL HARDWARE, OR AASHTO M 111, STANDARD SPECIFICATION FOR ZINC (HOT-DIP GALVANIZED) COATING ON IRON AND STEEL PRODUCTS (ASTM A123) FOR THE FITTINGS.
9. STREET LIGHTING POST CONCRETE FOOTING SHALL BE CLASS A AND SHALL CONFORM WITH SUBSECTION 405.2, MATERIAL REQUIREMENTS OF ITEM 405, STRUCTURAL CONCRETE.
10. ALL ELECTRICAL LIGHTING POST FOOTINGS WITH DIMENSIONS INDICATED IN THE PLANS SHALL BE REINFORCED CONCRETE AND SHALL CONFORM WITH THE REQUIREMENTS FOR CONCRETE STRUCTURES OF THIS SPECIFICATION. EXCAVATION AND BACKFILL FOR FOUNDATION INCLUDING DISPOSAL OF SURPLUS MATERIAL SHALL BE PROVIDED. EXCAVATED HOLES FOR CONCRETE FOOTINGS SHALL BE NEAT OR PROPERLY FROMED FREE FROM LOOSE MATERIALS WHEN CONCRETE IS PLACED.
7. ALL OTHER REQUIREMENTS SHALL CONFORM TO ITEM 624 - ROADWAY LIGHTING OF THE DPWH STANDARD SPECIFICATIONS FOR HIGHWAYS, BRIDGES, AND AIRPORTS, 2013 EDITION.

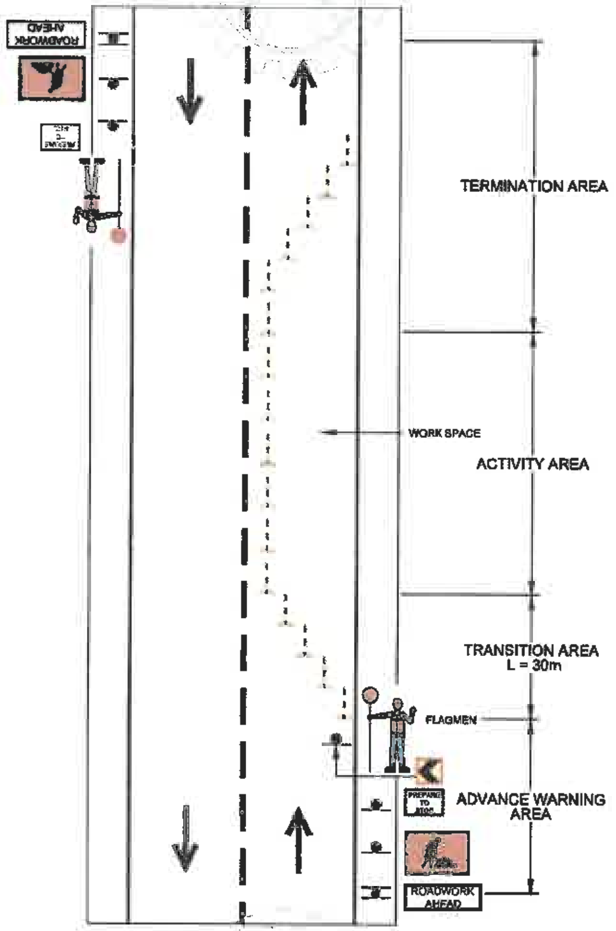
REPUBLIC OF THE PHILIPPINES DEPARTMENT OF PUBLIC WORKS AND HIGHWAYS REGION X CAGAYAN DE ORO CITY 1ST DISTRICT ENGINEERING OFFICE 10TH REGIONAL EQUIPMENT SERVICES COMPOUND BULUA, CAGAYAN DE ORO CITY	PROJECT NAME AND LOCATION:	SHEET CONTENTS:	SUBMITTED:	REVIEWED:	SUBMITTED:	RECOMMENDED:	APPROVED:	SET NO.:	SHEET NO.:
	CONCRETING OF ROAD WITH DRAINAGE MANTICAO DRIVE PHASE II, BRGY. CANITOAN, CAGAYAN DE ORO CITY, MISAMIS ORIENTAL	ROADWAY STREET LIGHTING	RONALD M. ABERIN ENGINEER II DRAFTED/PREPARED: [illegible] GRACE P. [illegible] ENGINEER II	ARGENT B. DAGALA ENGINEER II DATE	[illegible] MARCOS S. DACANAY CHIEF, [illegible] DESIGN SECTION DATE	EMILYNIA TIRADO ASSISTANT DISTRICT ENGINEER DATE	VERONICA [illegible] DATE	0 12 | 15	13 25

TYPICAL TRAFFIC SIGNAGE LAYOUT


ROADWORK AHEAD
PREPARE TO STOP
TERMINATION AREA
100' DOWNSTREAM TAPER
WORK SPACE
ACTIVITY AREA
BUFFER SPACE (LATERAL) 1.20mm MINIMUM
BUFFER SPACE (LONGITUDINAL)
TRANSITION AREA L = 30m
FLAGMEN
D
ADVANCE WARNING AREA
PREPARE TO STOP
ROADWORK AHEAD

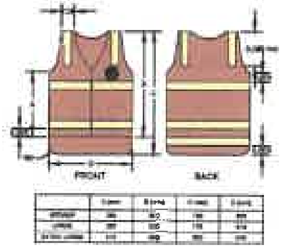
TYPICAL TRAFFIC SIGNAGES
(DURING DEMOLITION AND PAVING OPERATION)

SCALE: NTS


ROADWORK AHEAD
PREPARE TO STOP
TERMINATION AREA
WORK SPACE
ACTIVITY AREA
TRANSITION AREA L = 30m
FLAGMEN
ADVANCE WARNING AREA
PREPARE TO STOP
ROADWORK AHEAD

TYPICAL TRAFFIC SIGNAGES
(AFTER COMPLETION OF PAVING WORKS UP TO OPENING TO TRAFFIC)

SCALE: NTS



TRAFFIC SAFETY VEST DETAIL

NOTES :

1. ALL PERSONNEL WORKING ON OR ADJACENT TO ROADWORK SITE SHALL WEAR APPROPRIATE HIGH VISIBILITY CLOTHING.

2. THE VEST SHALL BE WORN OVER NORMAL CLOTHING AND PROPERLY FASTENED SO THAT THE ENTIRE AVAILABLE AREA OF HIGH VISIBILITY MATERIAL CAN BE SEEN IN ANY DIRECTION. TO MAXIMIZE EFFECTIVENESS THE VEST SHOULD BE KEPT CLEAN AND IN GOOD CONDITION.

3. THE TRAFFIC SAFETY VEST SHALL BE MADE FROM FLUORESCENT RED OR ORANGE MATERIAL. THE VEST SHALL ALSO HAVE TWO (2) STRIPS OF YELLOW RETRO-REFLECTIVE MATERIAL AT THE FRONT AND THREE (3) STRIPS AT THE BACK. REFLECTIVE MATERIAL SHALL BE 50mm WIDE. DPWH LOGO SHALL BE 80mm DIAMETER. THE SAFETY VEST SHOULD HAVE A SECURE FASTENING, PREFERABLY A ZIP.

4. THE JACKET MAY BE WORN OVER WET WEATHER CLOTHING. ALTERNATIVELY, WET WEATHER HIGH VISIBILITY CLOTHING SHALL BE MADE FROM WATER PROOF MATERIAL MATCHING THOSE PROPERTIES ABOVE FOR COLOR AND RETRO REFLECTIVITY.

NOTES :

1. ROADWORK AHEAD - THE FIRST ADVANCE WARNING SIGN SEEN BY DRIVERS, SHOULD BE LOCATED AT A MINIMUM DISTANCE "2D" METERS (D IS THE APPROACH SPEED OF TRAFFIC) BEFORE THE START OF THE WORKS, OR IF THERE IS A TRANSITION AREA, THE START OF THE TAPER.

2. OTHER SIGNS IN ADVANCE WARNING AREA SHOULD BE SPACED "D" METERS APART.

3. LONGITUDINAL BUFFER SPACE SHOULD BE 20 TO 30 METERS LONG BUT CAN BE EXTENDED IF THE WORK AREA IS HIDDEN FROM APPROACHING ROAD USERS, e.g. BY A CURVE OR CREST.

4. LATERAL BUFFER SPACE SHOULD BE A MINIMUM OF 1.2 METERS IN SLOW SPEED AREAS, OR IN HIGH SPEED AND HIGH VOLUME AREAS, A LARGE ADJACENT CLEARANCE IS DESIRABLE WITH SAFETY BARRIERS ADJACENT TO THE WORK AREA.

5. LOCATION OF TRAFFIC CONTROLLER SHOULD BE 30 METERS IN ADVANCE OF WORK AREA, WITH GOOD SITE DISTANCE OF ONCOMING TRAFFIC, WITH AN ESCAPE PATH IN AN EMERGENCY, WHERE THE OTHER TRAFFIC CONTROLLER IS VISIBLE, AND CAN BE SEEN BY THE ROAD USER.

REPUBLIC OF THE PHILIPPINES
DEPARTMENT OF PUBLIC WORKS AND HIGHWAYS
REGION X
CAGAYAN DE ORO CITY 1ST DISTRICT
ENGINEERING OFFICE
10TH REGIONAL EQUIPMENT SERVICES COMPOUND
BULUA, CAGAYAN DE ORO CITY

PROJECT NAME AND LOCATION:
CONCRETING OF ROAD WITH DRAINAGE
MANTICAO DRIVE PHASE II, BRGY.
CANITOAN, CAGAYAN DE ORO CITY,
MISAMIS ORIENTAL

SHEET CONTENTS:
TYPICAL TRAFFIC SIGNAGE LAYOUT

SURVEYED:
RONALD M. [illegible]
ENGINEER II

DRAFTED / [illegible]:
[illegible] GRACE P. [illegible]
ENGINEER II

REVIEWED:
[illegible] B. [illegible]ALA
ENGINEER II
DATE

SUBMITTED:
JO[illegible] B. DACA[illegible]
CHIEF, PLANNING AND DESIGN SECTION
DATE

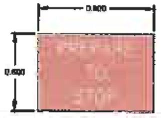
RECOMMENDED:
[illegible] A. TIRADO
ASSISTANT DISTRICT ENGINEER
DATE

APPROVED:
VIR[illegible]
DISTRICT ENGINEER
DATE

SET NO.: G 13 15

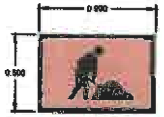
SHEET NO.: 14 25

TRAFFIC SIGNS AND HAZARD MARKERS


0.900
0.600
PREPARE
TO
STOP

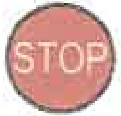
SIGN NO.	SIZE (mm)	LETTERS	BACKGROUND
T1-18	900 x 600	LINE 1 - WHITE 120 CM LINE 2 - WHITE 120 CM LINE 3 - WHITE 120 CM REFLECTORIZED	RED REFLECTORIZED

T1 - 18 DETAIL



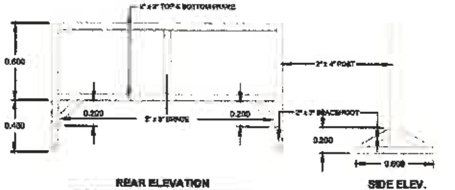
SIGN NO.	SIZE (mm)	LETTERS	BACKGROUND
T1-5	900 x 600	BLACK	RED/ORANGE - FLUORESCENT FOR DAY USE - REFLECTORIZED FOR NIGHT USE

T1 - 5 DETAIL



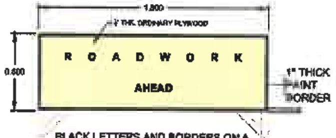
SIGN NO.	SIZE (mm)	LETTERS	BACKGROUND
R6-8	450 DIA.	WHITE 140 CM REFLECTORIZED (INCLUDING BORDER)	RED REFLECTORIZED

R6 - 8 DETAIL


2" x 2" TOP & BOTTOM FRAME
0.600
0.450
0.200
2" x 3" BRACE
0.200
2" x 4" POST
2" x 3" BRACE/FOOT
0.200
0.600

REAR ELEVATION
SCALE: NTS

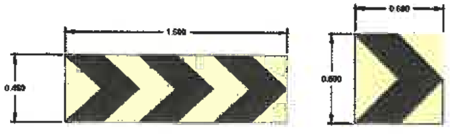
SIDE ELEV.
SCALE: NTS


1.800
1" THK. ORDINARY PLYWOOD
0.600
ROADWORK
AHEAD
1" THICK PAINT BORDER

BLACK LETTERS AND BORDERS ON A YELLOW PAINT W/ REFLECTORIZED STICKER
FRONT ELEVATION
SCALE: NTS

SIGN NO.	SIZE (mm)	LETTERS	BACKGROUND
T1-1	1800 x 600	LINE 1 - BLACK 200 CM LINE 2 - BLACK 150 CM	YELLOW REFLECTORIZED

T1 - 1 DETAIL

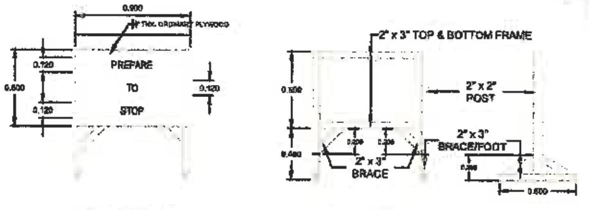

1.500
0.450
0.600
0.600

T5 - 4
SCALE: NTS

T5 - 5
SCALE: NTS

SIGN NO.	SIZE (mm)	CHEVRONS	BACKGROUND
T5-4	1500 x 450	BLACK 177 WIDE AT 450	YELLOW REFLECTORIZED
T5-5	400 x 600	BLACK 150 WIDE AT 450	YELLOW REFLECTORIZED

T5 - 4 & T5 - 5 DETAIL

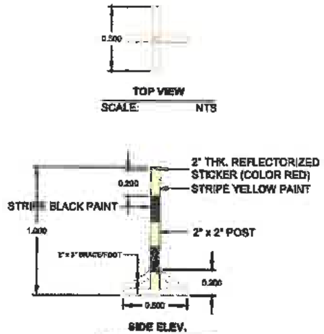

0.900
1" THK. ORDINARY PLYWOOD
0.120
0.600
0.120
0.120
PREPARE
TO
STOP
0.450
2" x 3" TOP & BOTTOM FRAME
0.600
2" x 2" POST
2" x 3" BRACE/FOOT
2" x 3" BRACE
0.200
0.200
0.450
0.600

FRONT ELEVATION
SCALE: NTS

REAR ELEVATION
SCALE: NTS

SIDE ELEV.
SCALE: NTS

TRAFFIC SIGNAGE INSTALLATION DETAIL


0.300
2" THK. REFLECTORIZED STICKER (COLOR RED)
STRIPE YELLOW PAINT
0.200
STRIPE BLACK PAINT
1.000
2" x 2" POST
2" x 3" BRACE/FOOT
0.200
0.500

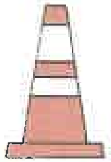
TOP VIEW
SCALE: NTS

SIDE ELEV.
SCALE: NTS

SPACING	CONDITION
5 - 10 M	ON TAPERS
	WHEN USED AROUND SMALL WORK SITES
	(MAY BE REDUCED TO 3m TO GUIDE PEDESTRIANS OR TO PREVENT TRAFFIC TAKING A WRONG TURN THROUGH A GAP IN THE LINE OF BOLLARDS)
10 - 20 M	WHEN USED AS LONGITUDINAL SEPARATION BETWEEN OPPOSING TRAFFIC FLOWS
	WHEN USED AS LONGITUDINAL SEPARATION OF TRAFFIC FROM THE WORKSITE OR A CLOSED LANE
	(MAY BE INCREASED TO 50m WHERE THE LENGTH OF BOLLARDS EXCEEDS 1 [illegible]

NOTES :
1. BOLLARDS SHALL BE FLUORESCENT RED OR ORANGE PLASTIC THAT IS RESILIENT TO IMPACT AND WILL NOT DAMAGE VEHICLES WHEN HIT AT LOW SPEED.
2. HEIGHT: UP TO 1 M.
3. AT NIGHT TIME OPERATION THE BOLLARDS MUST BE FITTED WITH REFLECTIVE TAPE WITH A MINIMUM BANDWIDTH OF 250mm.

TEMPORARY BOLLARDS DETAIL



SPACING	CONDITION
5 - 10 M	ON TAPERS
	WHEN USED AROUND SMALL WORK SITES
	(MAY BE REDUCED TO 3m TO GUIDE PEDESTRIANS OR TO PREVENT TRAFFIC TAKING A WRONG TURN THROUGH A GAP IN THE LINE OF CONES)
10 - 20 M	WHEN USED AS LONGITUDINAL SEPARATION BETWEEN OPPOSING TRAFFIC FLOWS
	WHEN USED AS LONGITUDINAL SEPARATION OF TRAFFIC FROM THE WORKSITE OR A CLOSED LANE
	(MAY BE INCREASED TO 50m WHERE THE LENGTH OF CONES EXCEEDS 1 KM)

NOTES :
1. TRAFFIC CONES SHALL BE FLUORESCENT RED OR ORANGE PLASTIC. IT MUST BE RESILIENT TO IMPACT AND WILL NOT DAMAGE VEHICLES WHEN HIT AT LOW SPEED.
2. HEIGHT VARIES FROM 450mm UP TO 750mm FOR USE ON EXPRESSWAYS OR IN VERY HIGH SPEED SITUATIONS.
3. TRAFFIC CONES SHALL BE DESIGNED TO BE STABLE IN WIND AND THE AIR TURBULENCE FROM PASSING TRAFFIC.
4. AT NIGHT TIME OPERATION THE CONES MUST BE FITTED WITH REFLECTIVE TAPE WITH A MINIMUM BANDWIDTH OF 150mm.

TYPICAL TRAFFIC CONE DETAIL

REPUBLIC OF THE PHILIPPINES
DEPARTMENT OF PUBLIC WORKS AND HIGHWAYS
REGION X
CAGAYAN DE ORO CITY 1ST DISTRICT
ENGINEERING OFFICE
10TH REGIONAL EQUIPMENT SERVICES COMPOUND
BULUA, CAGAYAN DE ORO CITY

PROJECT NAME AND LOCATION:
CONCRETING OF ROAD WITH DRAINAGE
MANTICAO DRIVE PHASE II, BRGY.
CANITOAN, CAGAYAN DE ORO CITY,
MISAMIS ORIENTAL

SHEET CONTENTS:
TRAFFIC SIGNS AND HAZARD MARKERS

SURVEYED:
ENGINEER II

DRAFTED / PREPARED:
[illegible] GRACE P. CAMPOS

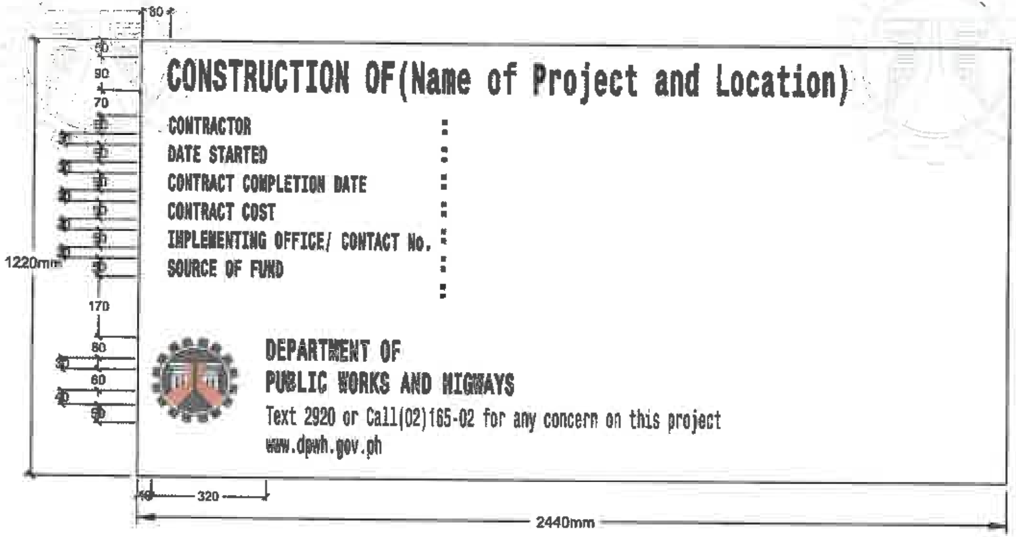
REVIEWED:
[illegible] R. DAGALA
DATE

SUBMITTED:
[illegible] E. DACANAY
[illegible] AND DESIGN SECTION
DATE

RECOMMENDED:
[illegible] A. TIRADO
[illegible] DISTRICT ENGINEER
DATE

APPROVED:
DISTRICT [illegible]
DATE

SET NO. : G 14 | 16
SHEET NO. : 18 / 25


80
90
90
70
1220mm
170
80
60
40
40
50
320
2440mm

CONSTRUCTION OF(Name of Project and Location)

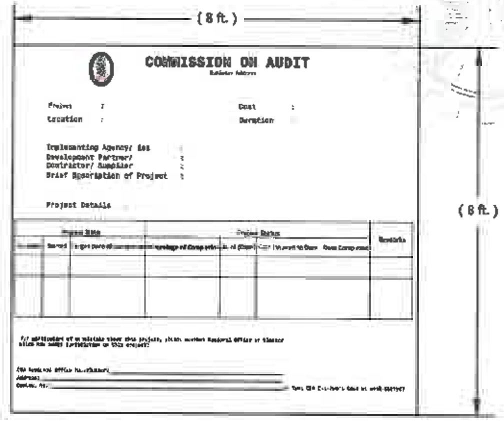
CONTRACTOR :
DATE STARTED :
CONTRACT COMPLETION DATE :
CONTRACT COST :
IMPLEMENTING OFFICE/ CONTACT No. :
SOURCE OF FUND :

DEPARTMENT OF
PUBLIC WORKS AND HIGWAYS
Text 2920 or Call(02)165-02 for any concern on this project
www.dpwh.gov.ph

SCALE 1 : 1 M.

NOTE :
For Source of Fund, state if DPWH Regular Budget, Priority Development Assistance Fund, DepEd/ DA/ DAR Budget, Calamity Fund, CVUC Fund, etc.

NOTE : THE EXPENSES INCURRED IN THE INSTALLATION OF BILLBOARD IS IN INCLUDED IN THE OCM PER DO. 12, SERIES 2011



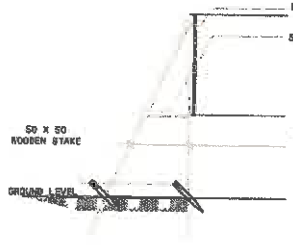
COMMISSION ON AUDIT

Project :
Location :
Cost :
Duration :

Implementing Agency/ ies :
Development Partner/ :
Contractor/ Supplier :
Brief Description of Project :

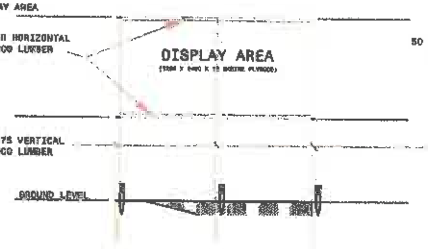
Project Details

Remarks

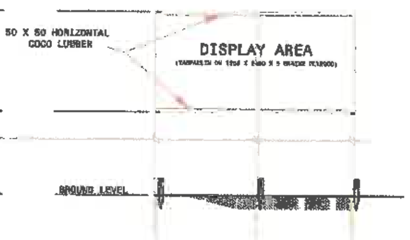

DISPLAY AREA
50 X 50 HORIZONTAL COCO LUMBER
50 X 50 WOODEN STAKE
50 X 75 VERTICAL COCO LUMBER
GROUND LEVEL

TYPICAL FRAME ELEVATION

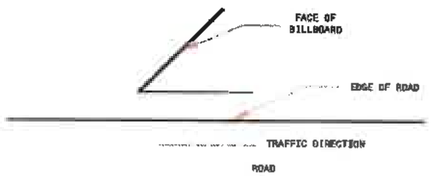
SCALE 1 : 25 M.


DISPLAY AREA
GROUND LEVEL

FRONT ELEVATION
(OPTION 1)


50 X 50 HORIZONTAL COCO LUMBER
DISPLAY AREA
GROUND LEVEL

FRONT ELEVATION
(OPTION 2)

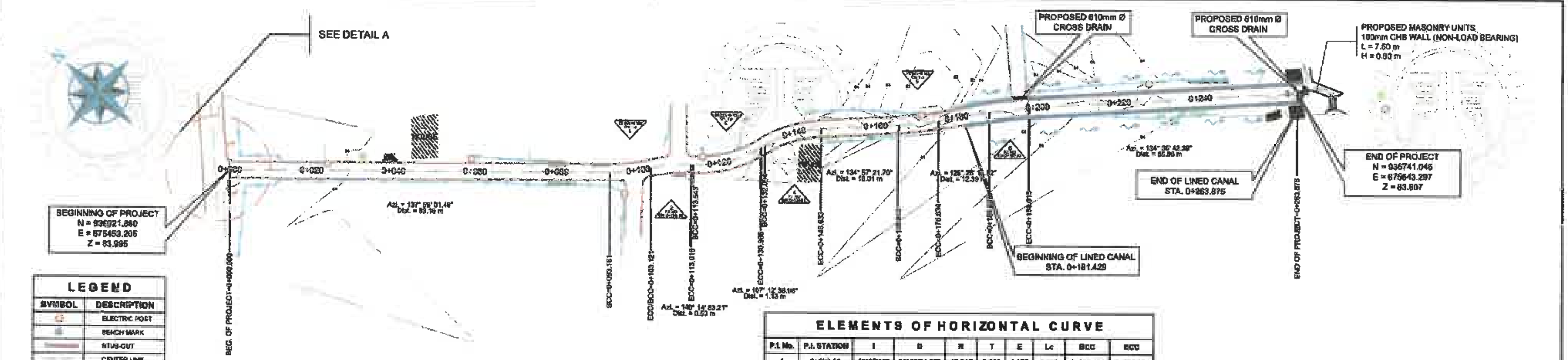

FACE OF BILLBOARD
EDGE OF ROAD
TRAFFIC DIRECTION
ROAD

ORIENTATION

STANDARD PROJECT BILLBOARD
(DPWH and COA)

REPUBLIC OF THE PHILIPPINES DEPARTMENT OF PUBLIC WORKS AND HIGHWAYS REGION X CAGAYAN DE ORO CITY 1ST DISTRICT ENGINEERING OFFICE 10TH REGIONAL EQUIPMENT SERVICES COMPOUND BULUA, CAGAYAN DE ORO CITY	PROJECT NAME AND LOCATION: CONCRETING OF ROAD WITH DRAINAGE MANTICAO DRIVE PHASE II, BRGY. CANITOAN, CAGAYAN DE ORO CITY, MISAMIS ORIENTAL	SHEET CONTENTS: PROJECT BILLBOARD	SURVEYED: ROMULO M. ABEPIN ENGINEER II DRAFTED/ JOREE GRACE A. SAMPOS ENGINEER II	REVIEWED: ALBERT B. DAGALA ENGINEER II DATE	SUBMITTED: JOSE [illegible] DACANAY CHIEF, PLANNING AND DESIGN SECTION DATE	RECOMMENDED: EVELYN A. TIRADO ASSISTANT DISTRICT ENGINEER DATE	APPROVED: VIRGINIA [illegible] DISTRICT ENGINEER DATE	SET NO. : G 15 | 16	SHEET NO. : 16 25

P. PLAN AND PROFILE


SEE DETAIL A
BEGINNING OF PROJECT
N = 936921.860
E = 675453.205
Z = 83.995
BEG. OF PROJECT=0+000.000
0+000
0+020
0+040
0+060
0+080
0+100
0+120
0+140
0+160
0+180
0+200
0+220
0+240
Azi. = 137° 59' 01.48"
Dist. = 83.76 m
BCC=0+093.161
ECC/BCC=0+103.121
ECC=0+113.016
BCC=0+113.543
Azi. = 140° 14' 53.21"
Dist. = 0.53 m
ECC=0+130.986
BCC=0+132.084
Azi. = 107° 12' 38.18"
Dist. = 1.33 m
ECC=0+146.633
Azi. = 134° 57' 21.70"
Dist. = 18.01 m
BCC=0+165.646
ECC=0+175.634
Azi. = 128° 26' 35.32"
Dist. = 12.39 m
BCC=0+188.025
ECC=0+198.015
BEGINNING OF LINED CANAL
STA. 0+181.429
PROPOSED 610mm Ø
CROSS DRAIN
PROPOSED 610mm Ø
CROSS DRAIN
Azi. = 134° 36' 42.38"
Dist. = 55.86 m
PROPOSED MASONRY UNITS
100mm CHB WALL (NON-LOAD BEARING)
L = 7.50 m
H = 0.93 m
END OF PROJECT
N = 936741.046
E = 675643.297
Z = 83.607
END OF LINED CANAL
STA. 0+263.875
END OF PROJECT=0+263.875

LEGEND

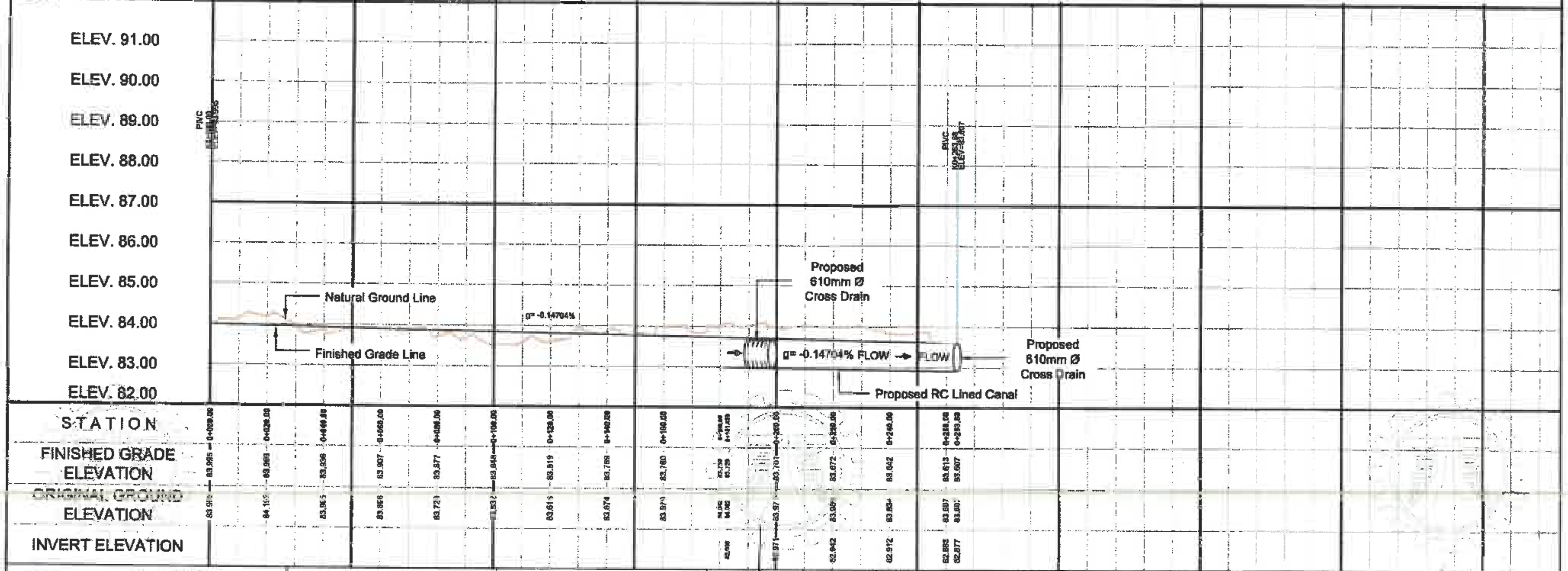
SYMBOL	DESCRIPTION
	ELECTRIC POST
	BENCH MARK
	STUB-OUT
	CENTER LINE
	ROAD EDGE
	FENCE
▬	CROSS DRAIN
	RAMP

ELEMENTS OF HORIZONTAL CURVE

P.I. No.	P.I. STATION	I	D	R	T	E	Lc	BCC	ECC
1	0+098.16	12°35'22"	38°39'31.87"	45.317	5.000	0.276	9.960	0+093.161	0+103.121
2	0+108.10	14°49'55"	46°12'54.10"	38.221	4.875	0.322	9.894	0+103.121	0+113.015
3	0+122.58	33°01'35"	59°30'15.01"	30.227	8.981	1.300	17.423	0+113.543	0+130.966
4	0+139.51	27°44'37"	59°56'30.05"	30.026	7.416	0.902	14.539	0+132.094	0+146.633
5	0+170.64	6°31'10"	19°40'34.42"	87.789	5.000	0.142	9.609	0+165.646	0+175.634
6	0+193.03	6°10'30"	19°37'31.67"	92.694	5.000	0.136	9.990	0+188.025	0+198.015

PLAN

1
P1
SCALE 1:700m


ELEV. 91.00
ELEV. 90.00
ELEV. 89.00
ELEV. 88.00
ELEV. 87.00
ELEV. 86.00
ELEV. 85.00
ELEV. 84.00
ELEV. 83.00
ELEV. 82.00
PIVC
PIVC
K0+263.88
ELEV=83.607
Natural Ground Line
Finished Grade Line
g= -0.14704%
Proposed
610mm Ø
Cross Drain
g= -0.14704% FLOW
FLOW
Proposed RC Lined Canal
Proposed
610mm Ø
Cross Drain

STATION	FINISHED GRADE ELEVATION	ORIGINAL GROUND ELEVATION	INVERT ELEVATION
0+000.00	83.995	83.995	
0+020.00	83.969	84.19[illegible]	
0+040.00	83.936	83.9[illegible]5	
0+060.00	83.907	83.856	
0+080.00	83.877	83.721	
0+100.00	83.848	83.53[illegible]	
0+120.00	83.819	83.61[illegible]	
0+140.00	83.789	83.674	
0+160.00	83.760	83.974	
0+180.00 0+181.429	83.739 83.729	84.062 84.062	82.896
0+200.00	83.701	83.97	[illegible].971
0+220.00	83.672	83.95[illegible]	82.942
0+240.00	83.642	83.83[illegible]	82.912
0+260.00 0+263.88	83.613 83.607	83.607 83.607	82.883 82.877

REPUBLIC OF THE PHILIPPINES
DEPARTMENT OF PUBLIC WORKS AND HIGHWAYS
REGION X
CAGAYAN DE ORO CITY 1ST DISTRICT
ENGINEERING OFFICE
10TH REGIONAL EQUIPMENT SERVICES COMPOUND
BULUA, CAGAYAN DE ORO CITY

PROJECT NAME AND LOCATION:
CONCRETING OF ROAD WITH DRAINAGE
MANTICAO DRIVE PHASE II, BRGY.
CANITOAN, CAGAYAN DE ORO CITY,
MISAMIS ORIENTAL

SHEET CONTENTS:
STA. 0+000.000 - STA. 0+253.875

SURVEYED:
ENGINEER II

DRAFTED BY:

REVIEWED:
ARDENT B. DAGALA
DATE

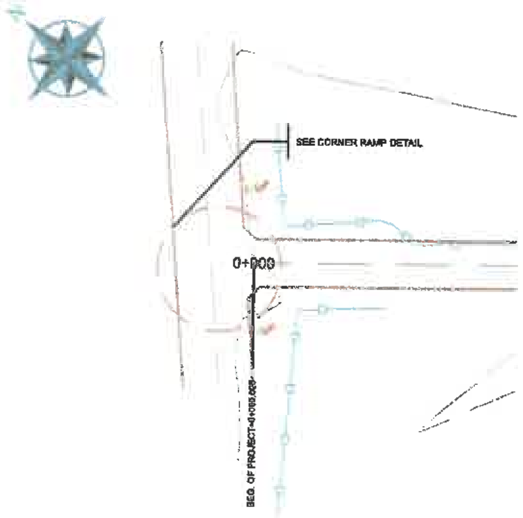
SUBMITTED:
JOSE [illegible] DACANAY
AND DESIGN SECTION
DATE

RECOMMENDED:
EVELYN A. TIRADO
DISTRICT ENGINEER
DATE

APPROVED:
VIR[illegible] R.
DATE

SET NO.: P 1 2

SHEET NO.: 17 / 25



DETAIL A

1
P 2

SCALE 1:3[illegible]

REPUBLIC OF THE PHILIPPINES
DEPARTMENT OF PUBLIC WORKS AND HIGHWAYS
REGION X
CAGAYAN DE ORO CITY 1ST DISTRICT
ENGINEERING OFFICE
10TH REGIONAL EQUIPMENT SERVICES COMPOUND
BULUA, CAGAYAN DE ORO CITY

PROJECT NAME AND LOCATION:
CONCRETING OF ROAD WITH DRAINAGE
MANTICAO DRIVE PHASE II, BRGY.
CANITOAN, CAGAYAN DE ORO CITY,
MISAMIS ORIENTAL

SHEET CONTENTS:
JUNCTION DETAILS

SURVEYED:
[illegible]
ENGINEER II

DRAFTED / [illegible]:
[illegible] GRACE [illegible]
[illegible]

REVIEWED:
AR[illegible] B. DIGALA
ENGINEER II
DATE

[illegible]:
JO[illegible] V. DACANAY
[illegible] AND DESIGN SECTION
DATE

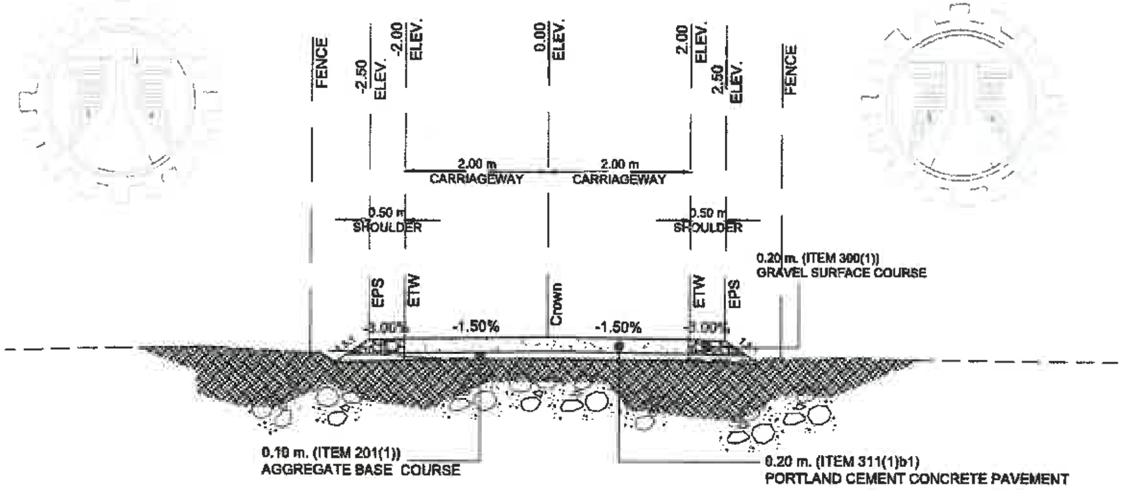
RECOMMENDED:
EV[illegible] TIRADO
ASS[illegible] DISTRICT ENGINEER
DATE

APPROVED:
VIR[illegible] IL
[illegible]
DATE

SET NO.:
P
2 2

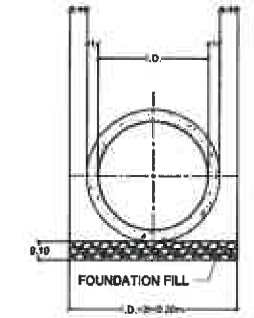
SHEET NO.:
18
25

S. TYPICAL SECTIONS, DETAILS AND QUANTITIES


FENCE
-2.50 ELEV.
-2.00 ELEV.
0.00 ELEV.
2.00 ELEV.
2.50 ELEV.
FENCE
2.00 m CARRIAGEWAY
2.00 m CARRIAGEWAY
0.50 m SHOULDER
0.50 m SHOULDER
EPS
ETW
Crown
ETW
EPS
-3.00%
-1.50%
-1.50%
-3.00%
0.20 m. (ITEM 300(1))
GRAVEL SURFACE COURSE
0.10 m. (ITEM 201(1))
AGGREGATE BASE COURSE
0.20 m. (ITEM 311(1)b1)
PORTLAND CEMENT CONCRETE PAVEMENT

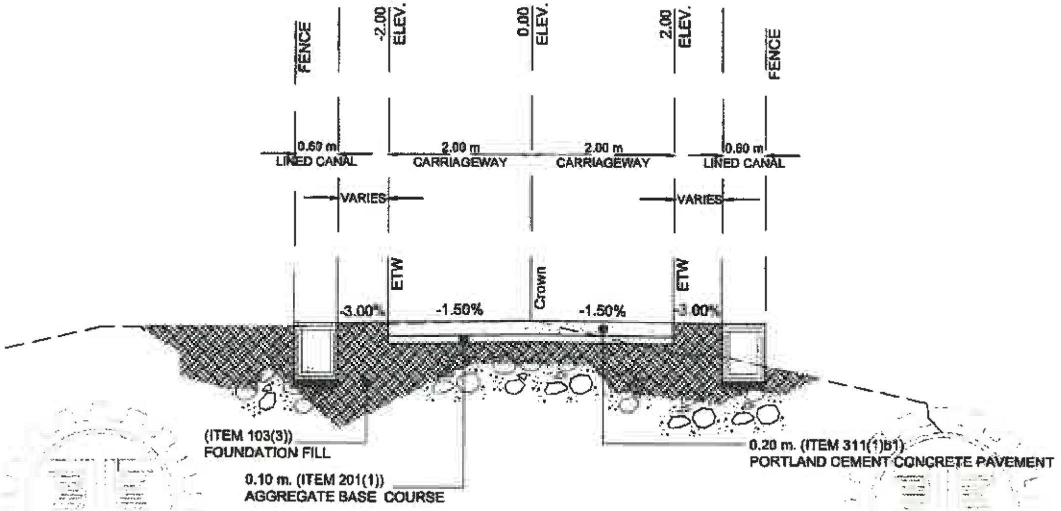
TYPICAL ROADWAY SECTION

1/S1 SCALE 1 : 5


I.D.
0.10
FOUNDATION FILL
I.D.+2+0.30m

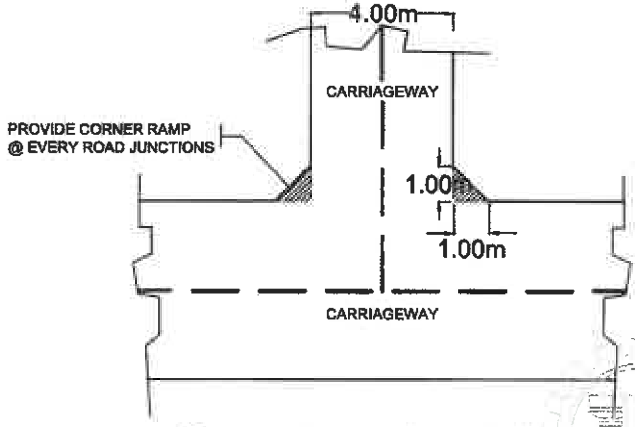
RCPC DETAILS

3/S1 SCALE 1 : 2


FENCE
-2.00 ELEV.
0.00 ELEV.
2.00 ELEV.
FENCE
0.60 m LINED CANAL
2.00 m CARRIAGEWAY
2.00 m CARRIAGEWAY
0.60 m LINED CANAL
VARIES
VARIES
ETW
Crown
ETW
-3.00%
-1.50%
-1.50%
-3.00%
(ITEM 103(3))
FOUNDATION FILL
0.10 m. (ITEM 201(1))
AGGREGATE BASE COURSE
0.20 m. (ITEM 311(1)b1)
PORTLAND CEMENT CONCRETE PAVEMENT

TYPICAL ROADWAY SECTION W/ LINED CANAL

2/S1 SCALE 1 : 5


4.00m
CARRIAGEWAY
PROVIDE CORNER RAMP
@ EVERY ROAD JUNCTIONS
1.00m
1.00m
CARRIAGEWAY

CORNER RAMP DETAIL

4/S1 SCALE 1 : 10

REPUBLIC OF THE PHILIPPINES
DEPARTMENT OF PUBLIC WORKS AND HIGHWAYS
REGION X
CAGAYAN DE ORO CITY 1ST DISTRICT
ENGINEERING OFFICE
10TH REGIONAL EQUIPMENT SERVICES COMPOUND
BULUA, CAGAYAN DE ORO CITY

PROJECT NAME AND LOCATION:
CONCRETING OF ROAD WITH DRAINAGE
MANTICAO DRIVE PHASE II, BRGY.
CANITOAN, CAGAYAN DE ORO CITY,
MISAMIS ORIENTAL

SHEET CONTENTS:
TYPICAL ROADWAY SECTIONS &
RCPC DETAILS

SURVEYED:
RONALD R. ASENJO
ENGINEER II

DRAFTED / [illegible]:
[illegible] P. [illegible]
ENGINEER II

REVIEWED:
ARDENT B. DAGALA
ENGINEER II
DATE

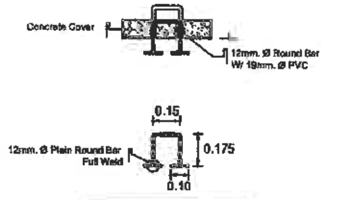
SUBMITTED:
JO[illegible] DACA[illegible]Y
[illegible] AND DESIGN SECTION
DATE

RECOMMENDED:
EVELYN A. TIRADO
ASSISTANT DISTRICT ENGINEER
DATE

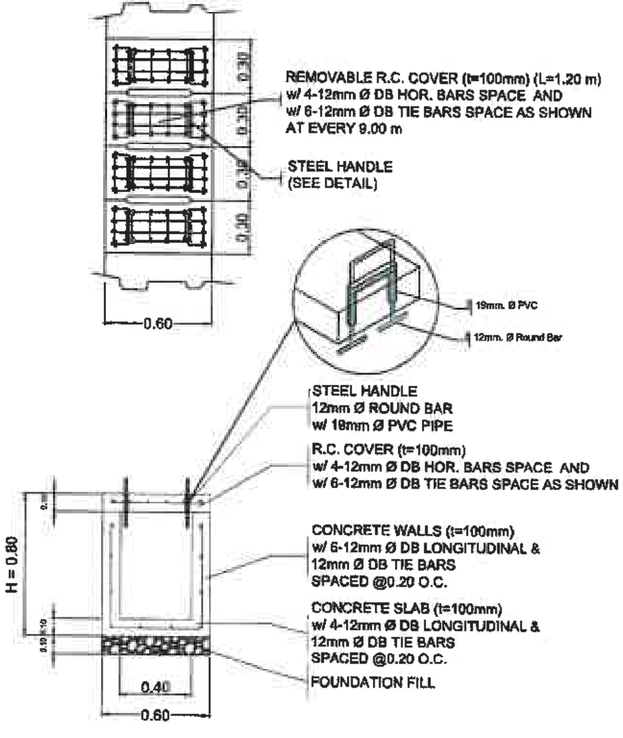
APPROVED:
VIRG[illegible]
DISTRICT ENGINEER
DATE

SET NO.: 5 / 1 3

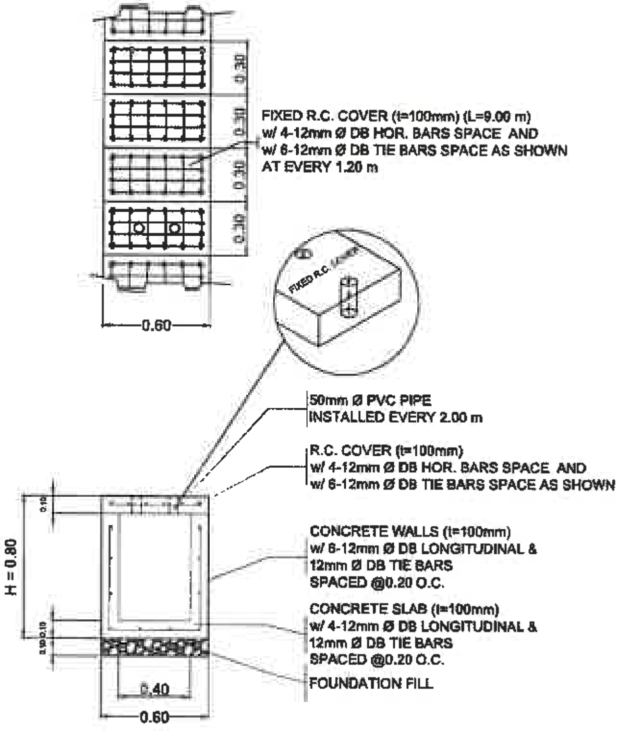
SHEET NO.: 19 / 25


Concrete Cover
12mm. Ø Round Bar W/ 19mm. Ø PVC
0.15
0.175
0.10
12mm. Ø Plain Round Bar Full Weld

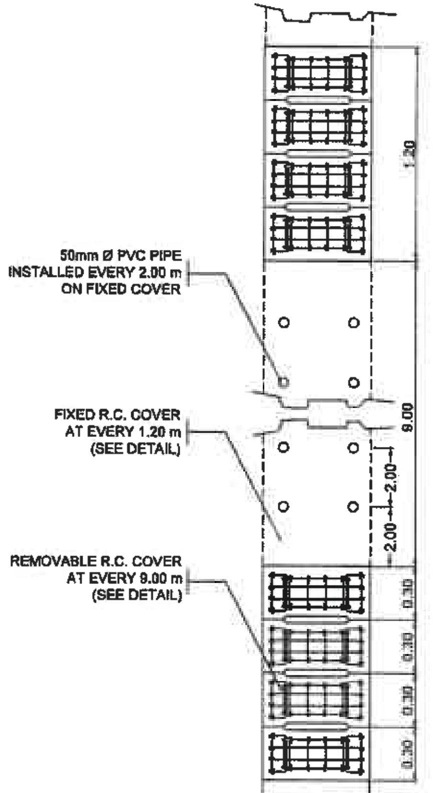
STEEL HANDLE DETAIL


0.30
0.30
0.30
0.30
0.60
REMOVABLE R.C. COVER (t=100mm) (L=1.20 m)
w/ 4-12mm Ø DB HOR. BARS SPACE AND
w/ 6-12mm Ø DB TIE BARS SPACE AS SHOWN
AT EVERY 9.00 m
STEEL HANDLE
(SEE DETAIL)
19mm. Ø PVC
12mm. Ø Round Bar
STEEL HANDLE
12mm Ø ROUND BAR
w/ 19mm Ø PVC PIPE
R.C. COVER (t=100mm)
w/ 4-12mm Ø DB HOR. BARS SPACE AND
w/ 6-12mm Ø DB TIE BARS SPACE AS SHOWN
CONCRETE WALLS (t=100mm)
w/ 6-12mm Ø DB LONGITUDINAL &
12mm Ø DB TIE BARS
SPACED @0.20 O.C.
CONCRETE SLAB (t=100mm)
w/ 4-12mm Ø DB LONGITUDINAL &
12mm Ø DB TIE BARS
SPACED @0.20 O.C.
FOUNDATION FILL
H = 0.80
0.10
0.10
0.10
0.40
0.60

TYPICAL SECTION WITH REMOVABLE COVER (R.C. CANAL)


0.30
0.30
0.30
0.30
0.60
FIXED R.C. COVER (t=100mm) (L=9.00 m)
w/ 4-12mm Ø DB HOR. BARS SPACE AND
w/ 6-12mm Ø DB TIE BARS SPACE AS SHOWN
AT EVERY 1.20 m
FIXED R.C. COVER
50mm Ø PVC PIPE
INSTALLED EVERY 2.00 m
R.C. COVER (t=100mm)
w/ 4-12mm Ø DB HOR. BARS SPACE AND
w/ 6-12mm Ø DB TIE BARS SPACE AS SHOWN
CONCRETE WALLS (t=100mm)
w/ 6-12mm Ø DB LONGITUDINAL &
12mm Ø DB TIE BARS
SPACED @0.20 O.C.
CONCRETE SLAB (t=100mm)
w/ 4-12mm Ø DB LONGITUDINAL &
12mm Ø DB TIE BARS
SPACED @0.20 O.C.
FOUNDATION FILL
H = 0.80
0.10
0.10
0.10
0.40
0.60

TYPICAL SECTION WITH FIXED COVER (R.C. CANAL)


1.20
50mm Ø PVC PIPE
INSTALLED EVERY 2.00 m
ON FIXED COVER
FIXED R.C. COVER
AT EVERY 1.20 m
(SEE DETAIL)
9.00
2.00
2.00
REMOVABLE R.C. COVER
AT EVERY 9.00 m
(SEE DETAIL)
0.30
0.30
0.30
0.30
0.60

TYPICAL DRAINAGE TOP VIEW

REPUBLIC OF THE PHILIPPINES
DEPARTMENT OF PUBLIC WORKS AND HIGHWAYS
REGION X
CAGAYAN DE ORO CITY 1ST DISTRICT
ENGINEERING OFFICE
10TH REGIONAL EQUIPMENT SERVICES COMPOUND
BULUA, CAGAYAN DE ORO CITY

PROJECT NAME AND LOCATION:	SHEET CONTENTS:	SURVEYED:	REVIEWED:	SUBMITTED:	RECOMMENDED:	APPROVED:	SET NO.:	SHEET NO.:
CONCRETING OF ROAD WITH DRAINAGE MANTICAO DRIVE PHASE II, BRGY. CANITOAN, CAGAYAN DE ORO CITY, MISAMIS ORIENTAL	TYPICAL SECTIONS OF REINFORCED CONCRETE CANALS, STEEL HANDLE DETAIL	[illegible] ENGINEER II DRAFTED: CHRIS [illegible]	ARSENIO [illegible] DAGALA ENGINEER II DATE	[illegible] DACANAY [illegible] AND DESIGN [illegible] DATE	[illegible]LYN A. TIRADO [illegible] DISTRICT ENGINEER DATE	VIR[illegible] [illegible] ENGINEER DATE	S 2 3	20 25

PAVEMENT DESIGN PARAMETERS

Design Life = 20 Years
Assumed Traffic Growth, TGR = 4.00 %
R = 85 %
ZR = - 1.037
Overall Standard Deviation, S0 = 0.35
W18 = x 10⁶ (18 KIP ESAL)
pt = 2.50
Present Serviceability Index, ΔPSI = 2.00
S'c = 751 psi
Cd = 1.00
J = 3.80
Ec = 3.605 x 10⁶ psi
k = pci

APPROXIMATE SLOPE RATIOS

HEIGHT OF SLOPE (M.)	NATURE OF MATERIAL	CUT SLOPE RATIO	FILL SLOPE RATIO
0.0 to 1.0	EARTH	4:1	4:1
1.1 to 2.0	EARTH	2:1	2:1
1.1 to 2.0	SOFT ROCK, CORAL SHALE, CEMENTED GR.	VARIES FROM 1/2:1 TO 1:1	2:1
1.1 to 2.0	HARD ROCK,	NATURE BREAK (DESIGN SLOPE) 1/4:1	2:1
Over 2.0	EARTH	1 ½:1	1 ½:1
Over 2.0	SOFT ROCK, CORAL SHALE, CEMENTED GR.	VARIES FROM 1/2:1 TO 1:1	1/2:1 to 1:1
Over 2.0	HARD ROCK	NATURE BREAK (DESIGN SLOPE) 1/2:1	1 ½: 1 to 1 ½:1

1. CUT/FILL SLOPE STAKE VARIES ON THE STABILITY OF MATERIAL

TABLE FOR SHOULDER SLOPE "S" (high shoulder)

e	S
3	3
4	3
5	2
6	1
7	0
8	1

1. ALGEBRAIC DIFFERENCE OF CROSS - OVER SHOULD NOT MORE THAN 7 %
2. FOR E GREATER THAN 7 % SHOULDER BE PAVED WITH CONCRETE, THICKNESS = 150 MM.

SUMMARY OF QUANTITIES

	DESCRIPTION OF WORKS TO BE DONE	QUANTITY	UNIT
Part B:	**Other General Requirements**		
B.5	Project Billboard / Signboard	1.00	each
B.7 (2)	Occupational Safety and Health Program	1.00	lumpsum
B.8 (1)	Traffic Management	2.33	month
B.9	Mobilization/Demobilization	1.00	lumpsum
Part C:	**Earthwork**		
100 (3) a1	Individual Removal of Trees (small, 150 - 300 mm. dia.)	5.00	each
101 (6)	Removal of Structures and Obstruction (Concrete)	20.00	cu.m.
102 (2)	Surplus Common Excavation	600.92	cu.m.
103 (3)	Foundation Fill	9.89	cu.m.
Part D:	**Subbase And Base Course**		
201 (1)	Aggregate Base Course	138.94	cu.m.
Part E:	**Surface Courses**		
300 (1)	Gravel Surface Course	47.68	cu.m.
311 (1) b1	Portland Cement Concrete Pavement (Unreinforced) (0.20 m thick, 14 days)	1,056.50	sq.m.
Part F:	**Bridge Construction**		
404 (1) a	Reinforcing Steel (Grade 40)	5,280.05	kg
405 (1) a2	Structural Concrete (20.68 MPa, Class A, 14 days)	39.74	cu.m.
Part G:	**Drainage and Slope Protection Structures**		
500 (1) a1	Pipe Culverts (610 mm Ø)	16.00	l.m.
Part I:	**Materials Details**		
704 (1)	Masonry Units	75.00	each

PAVING QUANTITIES

	LENGTH	WIDTH	AREA
	m	m	m^2
SECTION I STA. 0+000.000 STA. 0+263.875	263.875	4.00	1,056.50
TOTAL	263.875		1,056.50

GRADING QUANTITIES

		102 (2)	201 (1)	300 (1)
	DISTANCE	Surplus Common Excavation	Aggregate Base Course	Gravel Surface Course
SECTION I STA. 0+000.000 STA. 0+263.875	263.875	600.92	138.94	47.68
TOTAL	**263.875**	**600.92**	**138.94**	**47.68**

SCHEDULE OF DRAINAGE

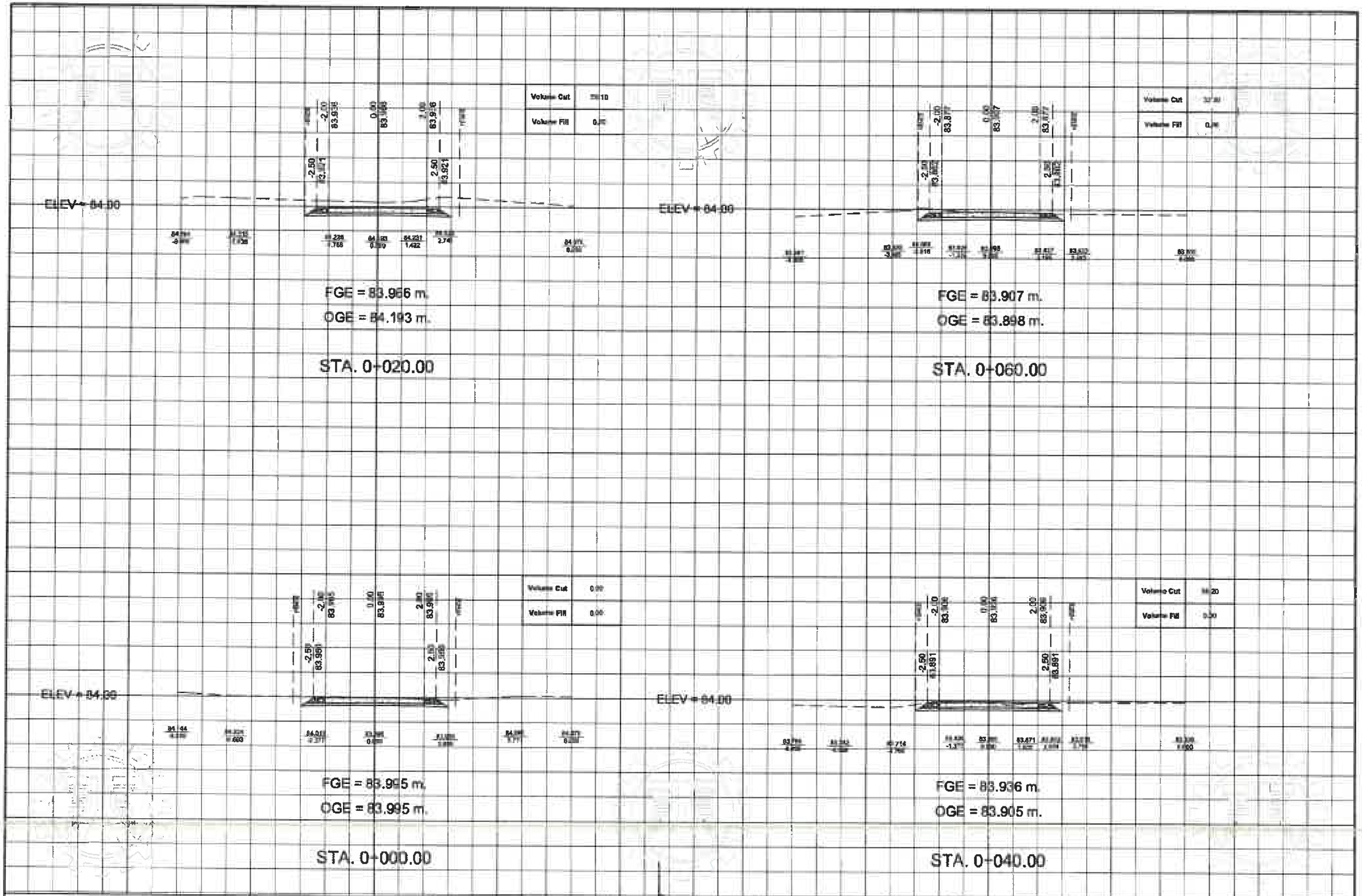
	COVER		WALLS		SLAB	
	HOR. BARS	TIE BARS	LONG. BARS	TIE BARS	LONG. BARS	TIE BARS
STA. 0+181.429 STA. 0+263.875	4-12 mm ø	6-12 mm ø	6-12 mm ø	12 mm ø, spaced @ 0.20 o.c.	4-12 mm ø	12 mm ø, spaced @ 0.20 o.c.

NOTE:

1. THE QUANTITIES OF ALL ITEMS OF WORK PRESENTED ARE SUBJECT TO INCREASE/ DECREASE DEPENDING ON THE ACTUAL ACCOMPLISHMENT AS CERTIFIED BY THE ENGINEER.
2. BEFORE THE START OF ACTUAL CONSTRUCTION, THE AS-STAKED PLAN SHOULD BE SUBMITTED TO THE DISTRICT OFFICE IN ORDER THAT IMMEDIATE STEP MAY BE UNDERTAKEN TO CORRECT OR ADJUST WHATEVER APPRECIABLE DEVIATION THERE MAY BE FROM THE ORIGINAL PLAN.
3. PRIOR TO CONSTRUCTION, SOIL INVESTIGATION SHALL BE CONDUCTED IN ACCORDANCE WITH DESIGN GUIDELINES AND TESTING REQUIREMENTS TO VERIFY AND VALIDATE DESIGN CBR.

REPUBLIC OF THE PHILIPPINES DEPARTMENT OF PUBLIC WORKS AND HIGHWAYS REGION X CAGAYAN DE ORO CITY 1ST DISTRICT ENGINEERING OFFICE 10TH REGIONAL EQUIPMENT SERVICES COMPOUND BULUA, CAGAYAN DE ORO CITY	PROJECT NAME AND LOCATION: CONCRETING OF ROAD WITH DRAINAGE MANTICAO DRIVE PHASE II, BRGY. CANITOAN, CAGAYAN DE ORO CITY, MISAMIS ORIENTAL	SHEET CONTENTS: SUMMARY OF QUANTITIES, GRADING QUANTITIES, PAVING QUANTITIES, SCHEDULE OF DRAINAGE	SURVEYED: RONALD [illegible] GEODETIC ENGINEER ENGINEER II DRAFTED / [illegible]: [illegible]	REVIEWED: [illegible] ENGINEER II DATE	SUBMITTED: [illegible] S. DACANAY [illegible] AND DESIGN SECTION DATE	RECOMMENDED: EVELYN A. TIRADO ASSISTANT DISTRICT ENGINEER DATE	APPROVED: VIR[illegible] DISTRICT ENGINEER DATE	SET NO.: 8 3 | 3 SHEET NO.: 21 25

C. CROSS SECTION


Volume Cut
Volume Fill
0.00
-2.50
83.921
-2.00
83.936
0.00
2.00
83.936
2.50
83.921
ELEV = 84.00
84.231
1.422
FGE = 83.966 m.
OGE = 84.193 m.
STA. 0+020.00
Volume Cut
Volume Fill
-2.00
83.877
2.00
83.877
ELEV = 84.00
FGE = 83.907 m.
OGE = 83.898 m.
STA. 0+060.00
Volume Cut
0.00
Volume Fill
0.00
-2.50
83.956
0.00
83.995
2.00
83.995
ELEV = 84.00
84.144
84.013
FGE = 83.995 m.
OGE = 83.995 m.
STA. 0+000.00
Volume Cut
Volume Fill
-2.50
83.891
-2.00
83.906
0.00
2.00
83.906
2.50
83.891
ELEV = 84.00
83.714
83.671
FGE = 83.936 m.
OGE = 83.905 m.
STA. 0+040.00

REPUBLIC OF THE PHILIPPINES
DEPARTMENT OF PUBLIC WORKS AND HIGHWAYS
REGION X
CAGAYAN DE ORO CITY 1ST DISTRICT
ENGINEERING OFFICE
10TH REGIONAL EQUIPMENT SERVICES COMPOUND
BULUA, CAGAYAN DE ORO CITY

PROJECT NAME AND LOCATION:

CONCRETING OF ROAD WITH DRAINAGE
MANTICAO DRIVE PHASE II, BRGY.
CANITOAN, CAGAYAN DE ORO CITY,
MISAMIS ORIENTAL

SHEET CONTENTS:

STA. 0+000.000 - STA. 0+060.000

SURVEYED:

ENGINEER II

DRAFTED:

IONNIE GRACE P.

REVIEWED:

ARDENT B. DAGALA
ENGINEER II
DATE

SUBMITTED:

JOVE MARCUS S. DACANAY
CHIEF, PLANNING AND DESIGN SECTION
DATE

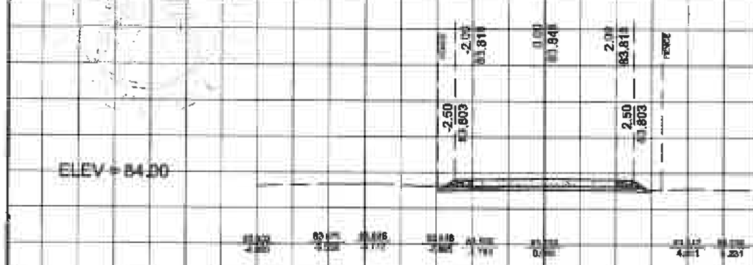
RECOMMENDED:

VIRADO
ASSISTANT DISTRICT ENGINEER
DATE

APPROVED:

VIRGINIA
DISTRICT ENGINEER
DATE

SET NO.:	SHEET NO.:
C 1 | 4	22 25

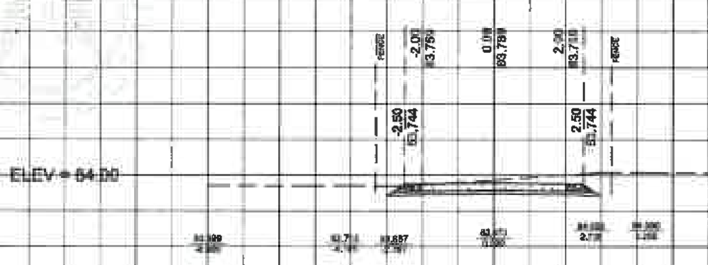

ELEV = 84.00
-2.50
83.803
-2.00
83.818
0.00
83.848
2.00
83.818
2.50
83.803
FENCE

Volume Cut	[illegible]
Volume Fill	[illegible]

FGE = 83.848 m.

OGE = 83.532 m.

STA. 0+100.00

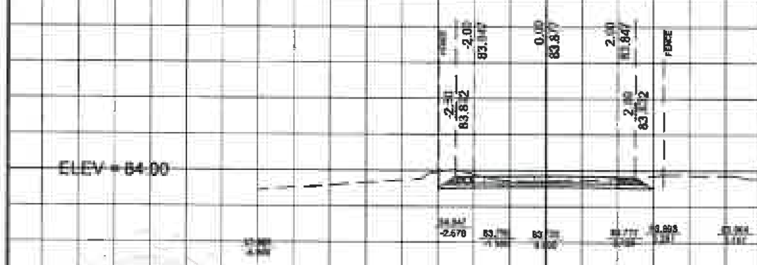

ELEV = 84.00
FENCE
-2.50
83.744
-2.00
83.759
0.00
83.789
2.00
83.759
2.50
83.744
FENCE

Volume Cut	30.30
Volume Fill	0.00

FGE = 83.789 m.

OGE = 83.873 m.

STA. 0+140.00

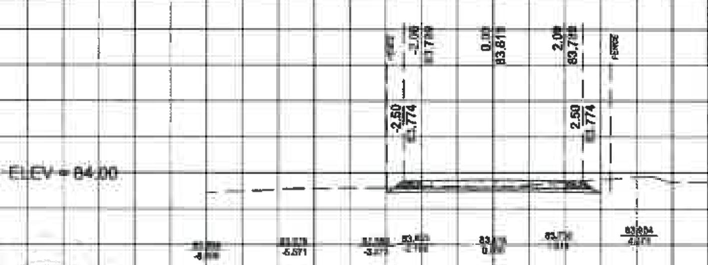

ELEV = 84.00
FENCE
-2.50
83.812
-2.00
83.847
0.00
83.877
2.00
83.847
2.50
83.812
FENCE

Volume Cut	[illegible]
Volume Fill	[illegible]

FGE = 83.877 m.

OGE = 83.720 m.

STA. 0+080.00


ELEV = 84.00
FENCE
-2.50
83.774
-2.00
83.789
0.00
83.819
2.00
83.789
2.50
83.774
FENCE

Volume Cut	[illegible]
Volume Fill	0.00

FGE = 83.819 m.

OGE = 83.616 m.

STA. 0+120.00

REPUBLIC OF THE PHILIPPINES
DEPARTMENT OF PUBLIC WORKS AND HIGHWAYS
REGION X
CAGAYAN DE ORO CITY 1ST DISTRICT
ENGINEERING OFFICE
10TH REGIONAL EQUIPMENT SERVICES COMPOUND
BULUA, CAGAYAN DE ORO CITY

PROJECT NAME AND LOCATION:
CONCRETING OF ROAD WITH DRAINAGE
MANTICAO DRIVE PHASE II, BRGY.
CANITOAN, CAGAYAN DE ORO CITY,
MISAMIS ORIENTAL

SHEET CONTENTS:
STA. 0+080.000 - STA. 0+140.000

SURVEYED:
ENGINEER II

DRAFTED:

REVIEWED:
ARDENT E. DAGALA
ENGINEER II
DATE

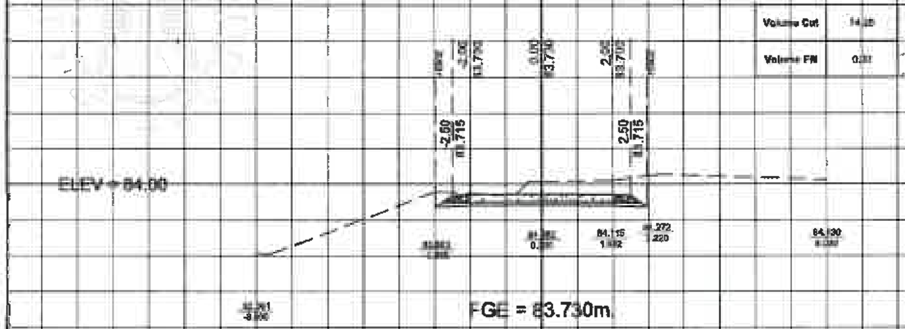
SUBMITTED:
DACANAY
AND DESIGN SECTION
DATE

RECOMMENDED:
A. TIRADO
ASSISTANT DISTRICT ENGINEER
DATE

APPROVED:
DISTRICT ENGINEER
DATE

SET NO.: C 2 4

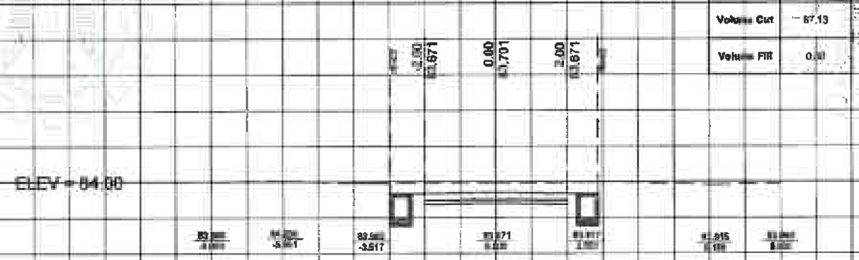
SHEET NO.: 23 / 25


ELEV = 84.00
-2.00
83.730
0.00
83.730
2.00
83.700
-2.50
83.715
2.50
83.715
84.115
1.982
84.272
.220
84.130

Volume Cut	[illegible]
Volume Fill	0.[illegible]

FGE = 83.730m.
OGE = 84.062 m.

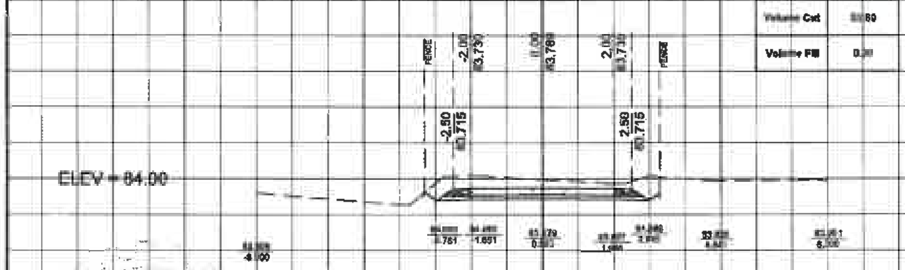
STA. 0+180.00


ELEV = 84.00
-2.00
83.671
0.00
83.701
2.00
83.671
83.5[illegible]
-3.517
[illegible].015

Volume Cut	67.13
Volume Fill	0.[illegible]

FGE = 83.701 m.
OGE = 83.971 m.
IE = 82.971 m.

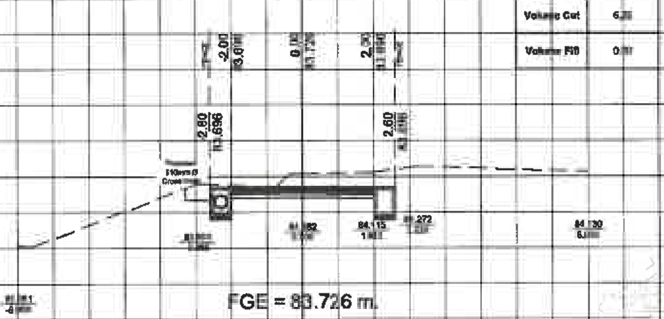
STA. 0+200.00


ELEV = 84.00
FENCE
-2.00
83.730
0.00
83.760
2.00
83.730
FENCE
-2.50
83.715
2.50
83.715
.781
-1.651
83.979
1.[illegible]
6.[illegible]

Volume Cut	[illegible]69
Volume Fill	0.[illegible]

FGE = 83.760 m.
OGE = 83.979 m.

STA. 0+160.00


ELEV = 84.00
-2.00
83.6[illegible]
0.00
83.72[illegible]
2.00
-2.60
83.696
2.60
910mm Ø
Cross
84.115
1.9[illegible]
84.272
84.130
6.[illegible]

Volume Cut	6.[illegible]
Volume Fill	0.[illegible]

FGE = 83.726 m.
OGE = 84.082 m.
IE = 82.[illegible] m.

STA. 0+181.429

REPUBLIC OF THE PHILIPPINES
DEPARTMENT OF PUBLIC WORKS AND HIGHWAYS
REGION X
CAGAYAN DE ORO CITY 1ST DISTRICT
ENGINEERING OFFICE
10TH REGIONAL EQUIPMENT SERVICES COMPOUND
BULUA, CAGAYAN DE ORO CITY

PROJECT NAME AND LOCATION:
CONCRETING OF ROAD WITH DRAINAGE
MANTICAO DRIVE PHASE II, BRGY.
CANITOAN, CAGAYAN DE ORO CITY,
MISAMIS ORIENTAL

SHEET CONTENTS:
STA. 0+160.000 - STA. 0+200.000

SURVEYED:
RONALD M. ABERIN
ENGINEER II

DRAFTED BY:
JOHNER GRACE P. CAMPOS
ENGINEER II

REVIEWED:
ARCENIO S. DAGALA
ENGINEER II
DATE

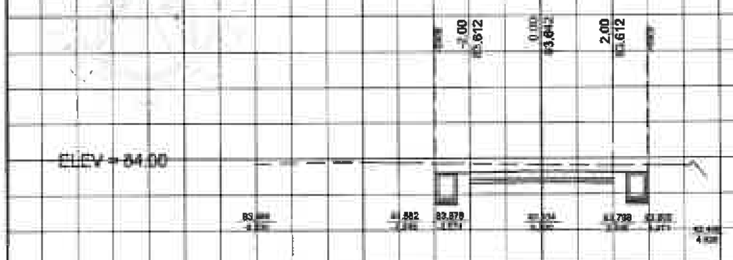
SUBMITTED:
[illegible] S. DACANAY
CHIEF, PLANNING AND DESIGN SECTION
DATE

RECOMMENDED:
EVELYN A. TIRADO
ASSISTANT DISTRICT ENGINEER
DATE

APPROVED:
VIR[illegible]
DATE

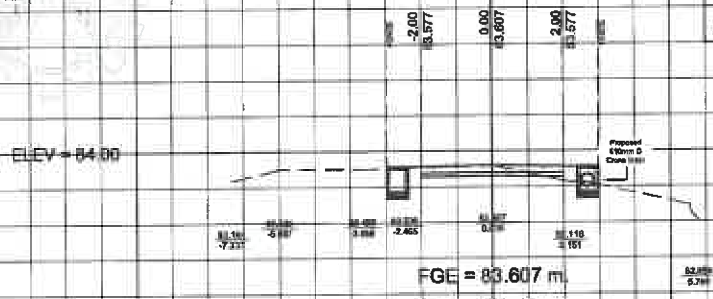
SET NO.: C 3 4

SHEET NO.: 24 25


ELEV = 84.00
-2.00
83.612
0.00
83.642
2.00
83.612

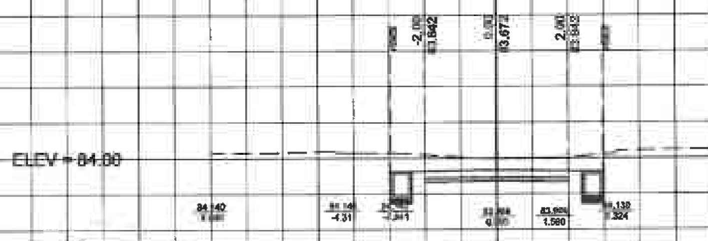
Volume Cut	[illegible]
Volume Fill	0.00

FGE = 83.642 m.
OGE = 83.834 m.
IE = 82.912 m.
STA. 0+240.00


ELEV = 84.00
-2.00
83.577
0.00
83.607
2.00
83.577
Proposed

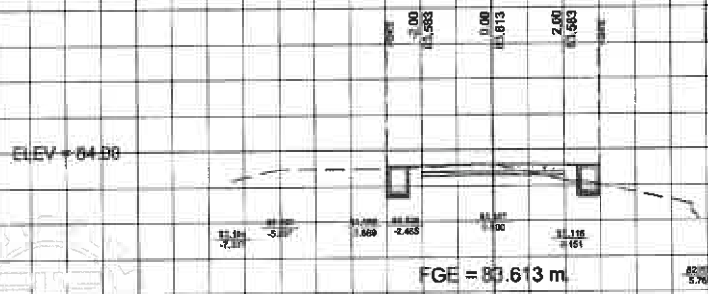
Volume Cut	8.14
Volume Fill	[illegible]

FGE = 83.607 m.
OGE = 83.607 m.
IE = 82.877 m.
STA. 0+263.875


ELEV = 84.00
-2.00
83.642
0.00
83.672
2.00
83.842

Volume Cut	[illegible]
Volume Fill	[illegible]

FGE = 83.672 m.
OGE = 83.989 m.
IE = 82.942 m.
STA. 0+220.00


ELEV = 84.00
-2.00
83.583
0.00
83.813
2.00
83.583

Volume Cut	[illegible]
Volume Fill	[illegible]

FGE = 83.613 m.
OGE = 83.607 m.
IE = 82.883 m.
STA. 0+260.00

REPUBLIC OF THE PHILIPPINES
DEPARTMENT OF PUBLIC WORKS AND HIGHWAYS
REGION X
CAGAYAN DE ORO CITY 1ST DISTRICT
ENGINEERING OFFICE
10TH REGIONAL EQUIPMENT SERVICES COMPOUND
BULUA, CAGAYAN DE ORO CITY

PROJECT NAME AND LOCATION:
CONCRETING OF ROAD WITH DRAINAGE
MANTICAO DRIVE PHASE II, BRGY.
CANITOAN, CAGAYAN DE ORO CITY,
MISAMIS ORIENTAL

SHEET CONTENTS:
STA. 0+220.000 - STA. 0+263.875

RONALD M. ABETON
ENGINEER II

JOHN DE GRACE F. CAMPOS

REVIEWED:
ARDENT B. DAGALA
DATE

SUBMITTED:
JOSE [illegible] F. DACANAY
AND DESIGN

RECOMMENDED:
[illegible] TIRADO
DISTRICT ENGINEER
DATE

APPROVED:
VIR[illegible]
DISTRICT ENGINEER
DATE

SET NO.: C 4 4
SHEET NO.: 25 25